

Nonstationary relationships in ecological modeling

Megan L. Feddern^{1,3}, Michael A. Litzow², Eric J. Ward³, Emily R. Ryznar², Mary E. Hunsicker⁴, Erin V. Satterthwaite⁵, Catarina Wor⁶, Sean C. Anderson^{6,7}, William H. Satterthwaite⁸, Michael J. Malick⁹, Dan A. Greenberg⁶, Mike G. Jacox^{10,11}, Brendan M. Connors^{12,13}

¹Fishery Resource Analysis and Monitoring Division, Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, 2725 Montlake Blvd E, Seattle, WA, USA

²Alaska Fisheries Science Center, National Marine Fisheries Service, Kodiak, AK USA

³Conservation Biology Division, Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, WA USA

⁴Fish Ecology Division, Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Newport, OR, USA

⁵California Sea Grant, Scripps Institution of Oceanography, University of California, San Diego, CA, United States of America

⁶Pacific Biological Station, Fisheries and Oceans Canada, Nanaimo, BC, Canada

⁷School of Resource and Environmental Management, Simon Fraser University, Burnaby, BC, Canada

⁸Fisheries Ecology Division, Southwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Santa Cruz, CA, USA

⁹Environmental and Fisheries Sciences Division, Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Port Orchard, WA, USA

¹⁰Ecosystem Science Division, Southwest Fisheries Science Center, National Oceanic and Atmospheric Administration, Monterey, CA, USA

¹¹Physical Sciences Laboratory, National Oceanic and Atmospheric Administration, Boulder, CO, USA

¹²Institute of Ocean Sciences, Fisheries and Oceans Canada, Sidney, BC, Canada

¹³Department of Biological Sciences, Simon Fraser University, Burnaby, BC, Canada

Corresponding Author: Megan L. Feddern, megan.feddern@noaa.gov

Appendix 1: Trout Lake

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Overview

Here we show the LM, and GAM analysis for the Trout Lake dataset. For complete code, see the [full quarto document](#)

Approach for running LTER-NTL Trout Lake data using temporally structured GAMs and linear regression models with time periods defined a-priori.

First we start with looking at three time periods using linear regression models:

1. The historical regime with low water clarity
2. A clear water regime where the introduction of Lake Trout into the system from stocking in 2006.
3. A novel regime following the introduction of invasive, predatory water flea (*Bythotrephes*) in 2014 which lead to a reversion of water clarity to a less clear state.

To examine this we start by looking at a few key food web conditions using intercept only models through time:

1. Water clarity
2. Phosphorus - which impacts water clarity and is a common bottom-up process that could impact water clarity and we examine as an alternative hypothesis to the top down processes of Lake Trout and invasive speceis.
3. Abundance of large zooplankton *Daphnia* and *Calanoids*
4. Chlorophyll

LM

We approach this by fitting a linear model with the *a priori* time periods as a factor and compare AIC to a single intercept model for each variable.

We find that water clarity, chlorophyll, and large zooplankton abundance are all better explained by a model that includes *a priori* defined time periods improves model fit (Table S1.1). This indicates that the means for these variables change through time. By examining the time series plot (Figure S1.1) we can see that large zooplankton and water clarity appear more correlated after 2006, when Lake Trout were stocked in the system. This aligns with our hypothesis that the introduction of top down control in the system has made large zooplankton abundance more tightly coupled with water clarity.

Next we consider whether there are changing relationships between water clarity and these ecosystem dynamics by including a slope parameter and its interaction with time period. We only include an investigation into large zooplankton-water clarity relationship in the main text, but here we also explore chlorophyll-water clarity.

Both chlorophyll and large zooplankton best explain water clarity with a nonstationary relationship model, rather than a time-dependent abundance model. We can look at the model residuals to see how they compare for a time-dependent model versus a time-independent model. Using the large zooplankton covariate as an example, we can see in the timeseries of residuals (Figure S1.2) that there is a temporal trend, supporting a temporally structured model. We find the q-q plot is similar across the models (Figure S1.3) and are mostly okay.

Next, we examine how these relationships change through time, with a focus on large zooplankton.

We can see from these results that there was a weak relationship from 1980 - 2006 between zooplankton abundance and water clarity, that switched to a strong, positive relationship in 2007 - 2014 as large zooplankton were released from top down pressure, and a slightly stronger relationship after 2014 with the introduction of *Bythotrephes* (Figure S1.4 & Table S1.3).

GAM

If we take the Zooplankton-water quality relationship and fit a temporally structured GAM and compare it to the linear model, we find similar results. Both predictions also accurately identify 2006 as a breakpoint and a second break point in 2016, although the GAM indicates that the slope of this relationship is a continuously changing trend (Figure S1.8) and the intercept more closely tracks abundance (Figure S1.9).

We can do similar residual diagnostics with the GAM model used. We can see that in a model that does not include a smooth by year term, there is a temporal trend in the residuals, but this is resolved when a smooth by year is implemented (Figure S1.6). There doesn't appear to be issues with the QQ plots for any specification (Figure S1.7).

DLM

For completeness, we also show the slope estimates and model fits for a DLM. We discuss the approach for a DLM in greater detail in Appendix 2: Lake Washington Analysis. Briefly, we find that all three modeling approaches have similar slope estimates (Figure S1.10) and fitted values (Figure S1.11) using all three modeling approaches.

Figures & Tables

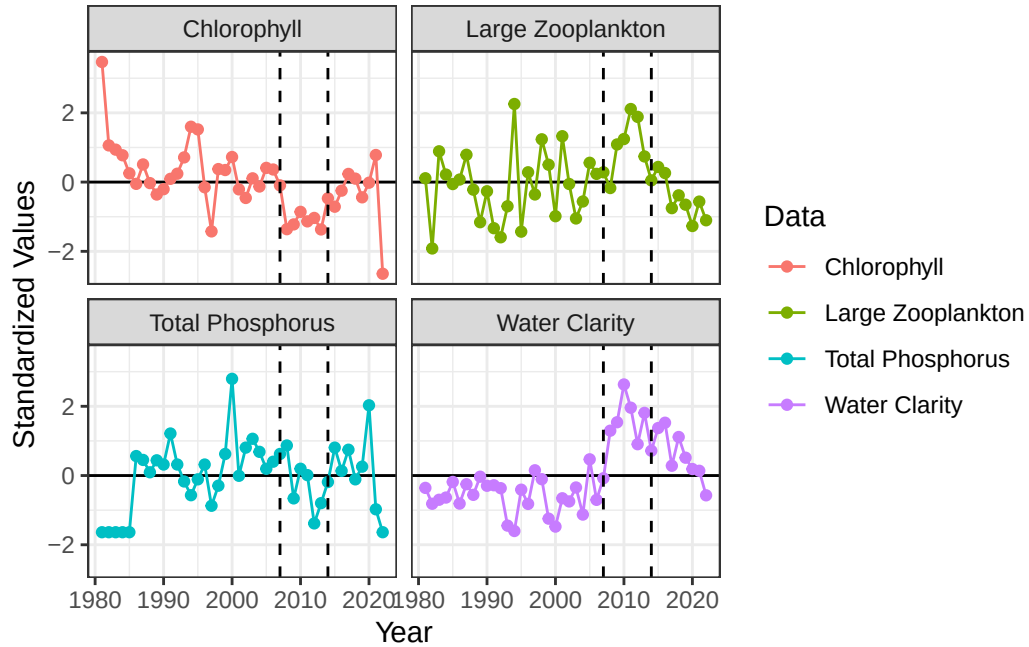


Figure S1.1: Standardized time series of ecological conditions observed in Trout Lake.

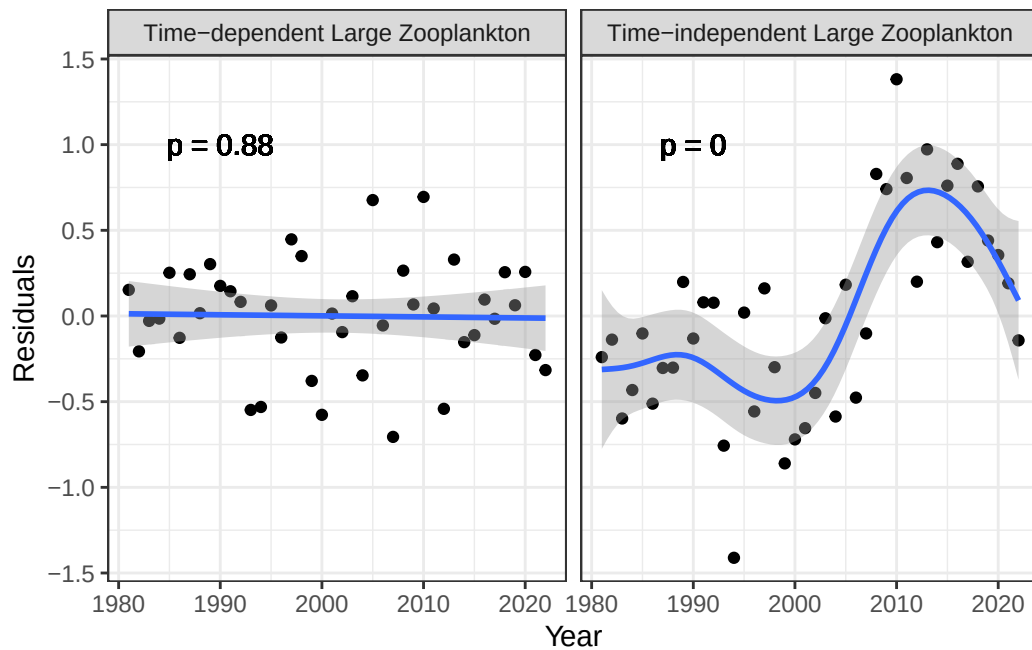


Figure S1.2: Model residuals for the time-dependent water clarity - zooplankton model compared to a time-independent model where large zooplankton is used as a covariate without *a priori* defined time periods. p-value denotes the significance of the trend in residuals through time.

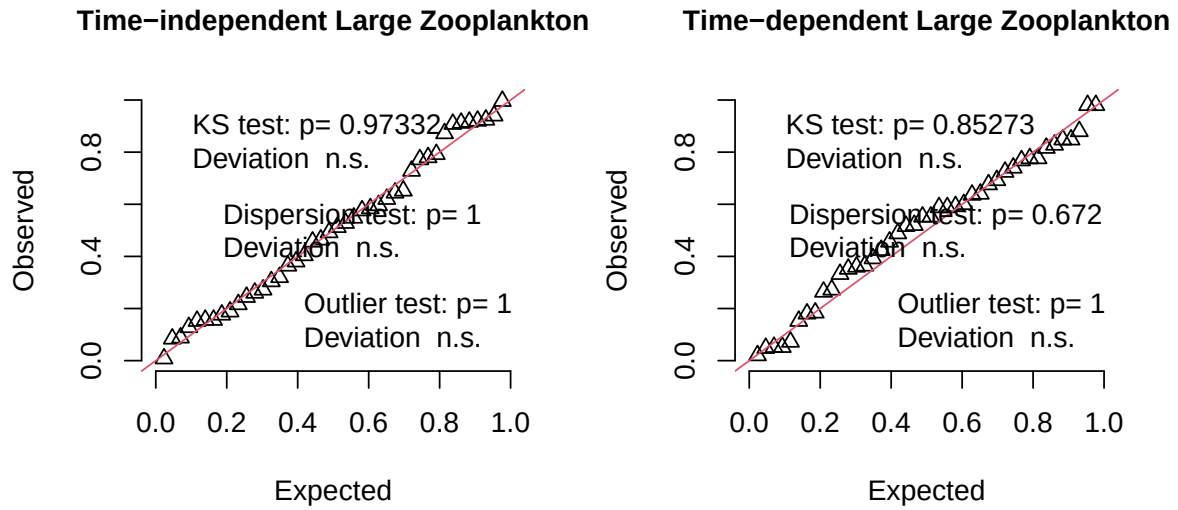


Figure S1.3: Q-Q plot for the time-dependent water clarity - zooplankton model compared to a time-independent model where large zooplankton is used as a covariate without *a priori* defined time periods

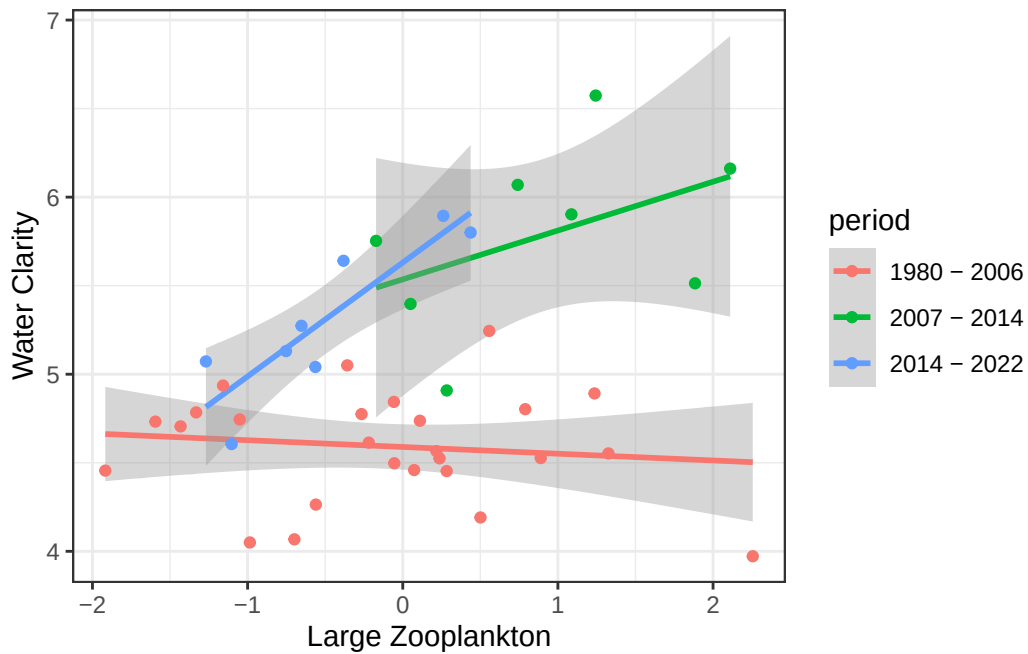


Figure S1.4: Relationship between large zooplankton abundance and water clarity for three time periods defined *a priori*.

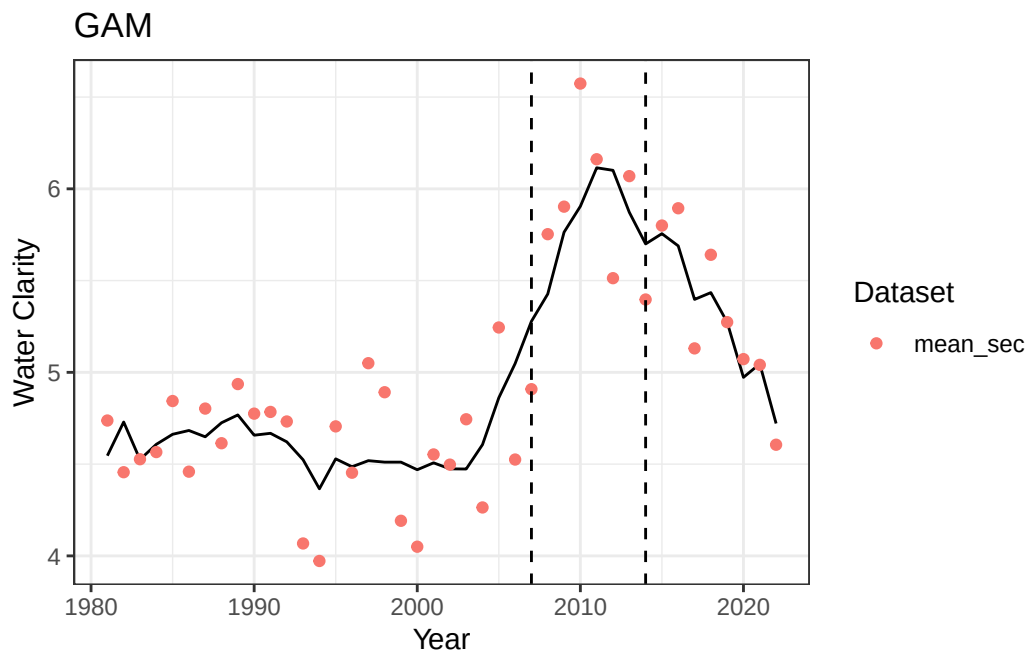


Figure S1.5: Predicted values of water clarity (black line) compared to observed values (red points) using a time-dependent relationship with zooplankton abundance.

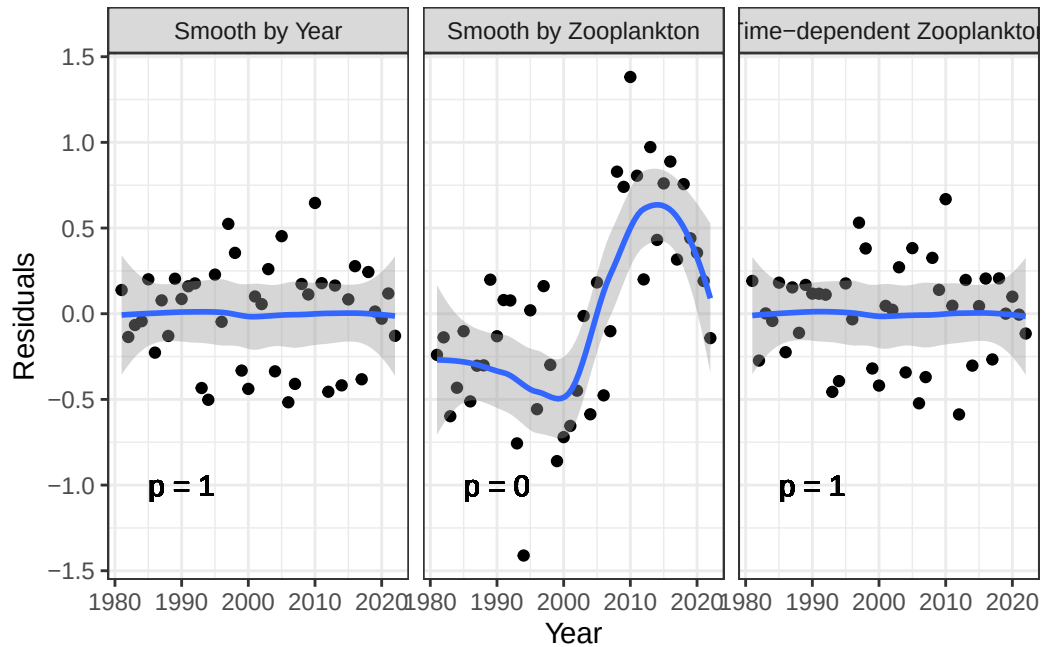


Figure S1.6: Model residuals for a smooth on zooplankton abundance, a smooth on year, and a smooth on year by zooplankton. p-value denotes the significance of the trend in residuals through time.

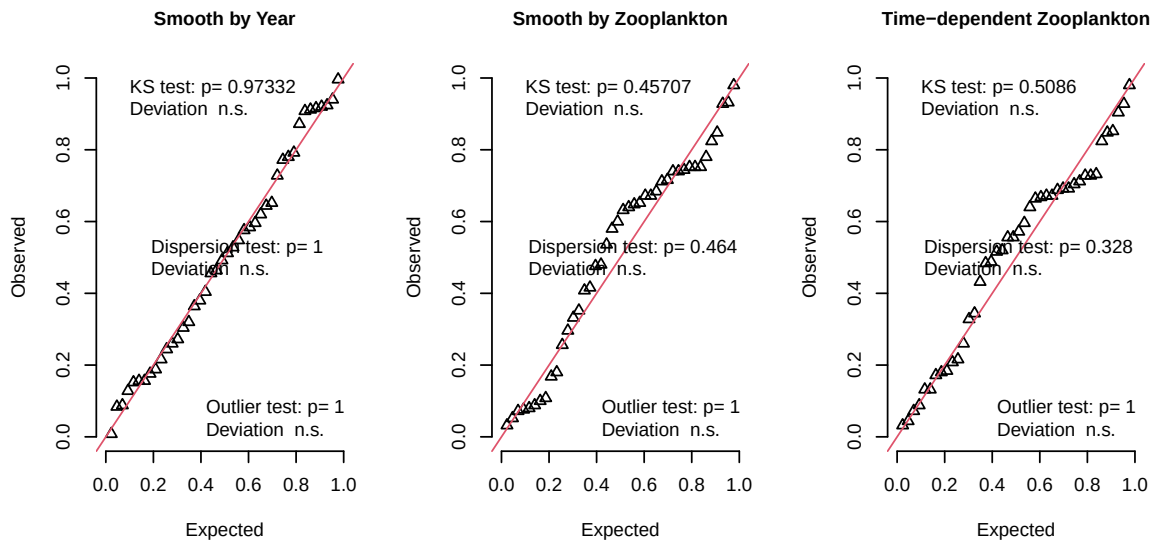


Figure S1.7: Q-Q plot for a smooth on zooplankton abundance, a smooth on year, and a smooth on year by zooplankton.

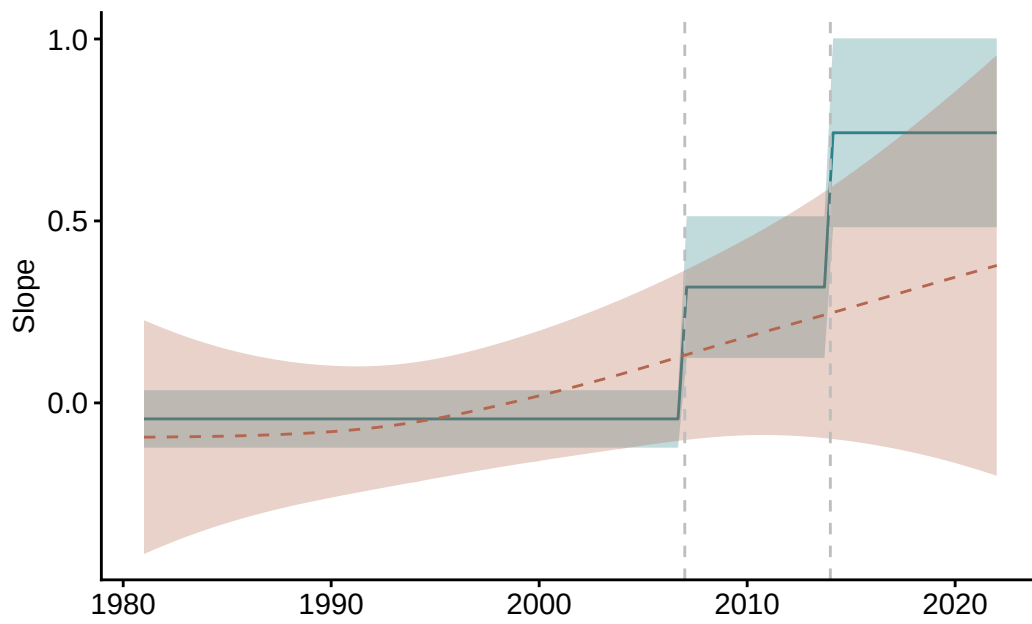


Figure S1.8: Estimates of the slope of the relationship based on a time-dependent LM (blue) and a time-dependent GAM (red)

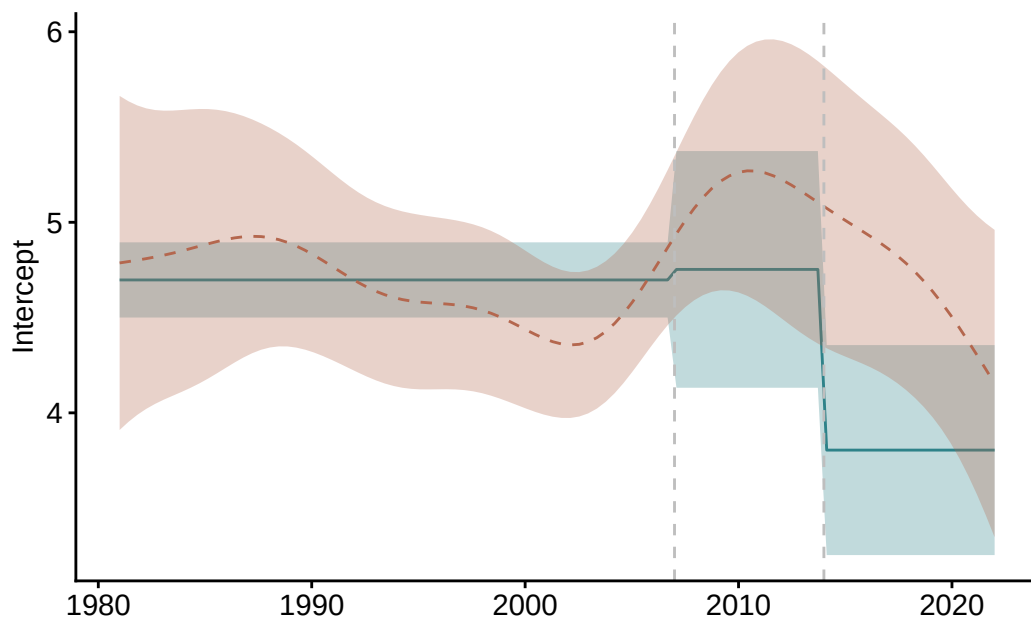


Figure S1.9: Estimates of the intercept of the relationship based on a time-dependent LM (blue) and a time-dependent GAM (red)

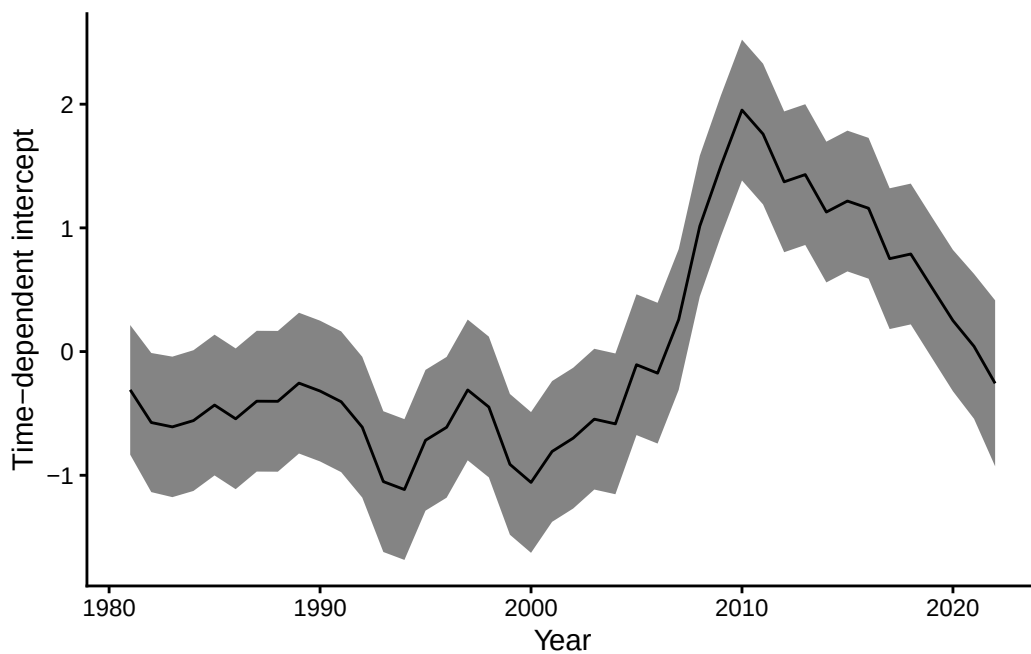


Figure S1.10: Estimates of the intercept of the relationship based on a time-dependent DLM

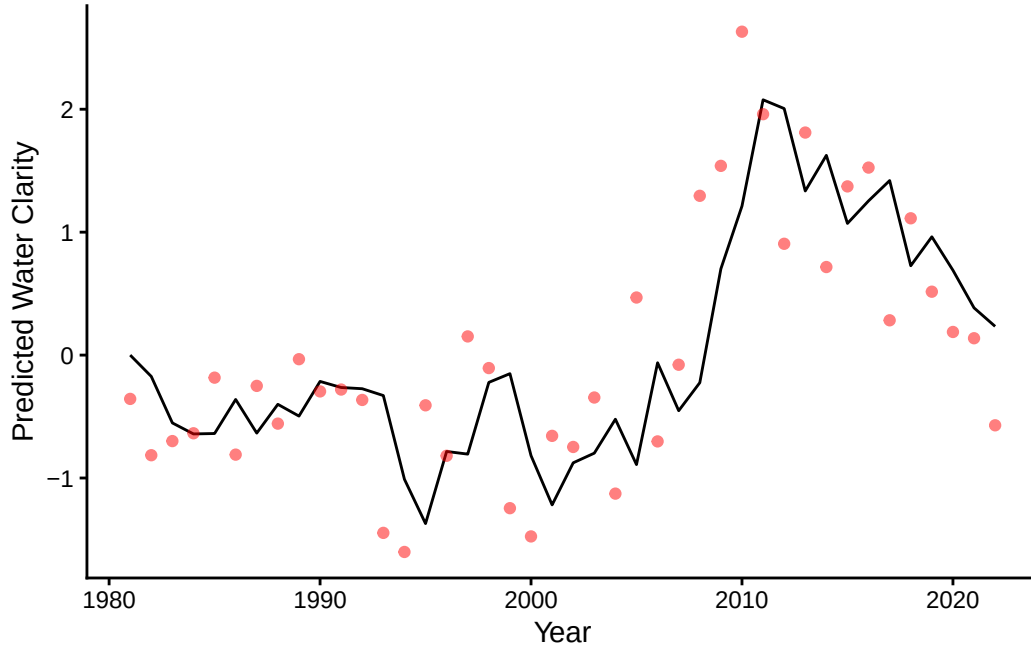


Figure S1.11: Predicted values of water clarity (black line) compared to observed values (red points) using a time-dependent relationship with zooplankton abundance fit with a DLM

Variable	No.Period.AIC	Period.AIC	Best.Model
Water Clarity	122.18	83.43	Period
Total Phosphorus	122.18	125.75	No difference
Chlorophyll	122.18	111.00	Period
Large Zooplankton	122.18	115.84	Period

Table S1.1: AIC values comparing linear models with *a priori* defined time periods (time-dependent model) and no time period (time-independent model)

Covariates	Period	Interaction
Chlorophyll	43.99	39.56
Total Phosphorus	44.45	46.31
Large Zooplankton	43.58	35.99

Table S1.2: AIC values comparing linear models comparing time-dependent abundance models versus nonstationary relationship models

term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.70	0.20	23.80	0.00
Large	-0.04	0.08	-0.56	0.58
as.factor(period)2	0.06	0.62	0.09	0.93
as.factor(period)3	-0.89	0.55	-1.62	0.11
Large:as.factor(period)2	0.36	0.19	1.86	0.07
Large:as.factor(period)3	0.79	0.26	3.03	0.00

Table S1.3: Model coefficients for the nonstationary relationship model between large zooplankton and water clarity

Appendix 2: Lake Washington Analysis

Appendix 2: Lake Washington Analysis

Here we show the GAM and DLM analysis for the Lake Washington dataset. For complete code, see the [full quarto document](#). We include some code snippets here for reference.

Data Processing

In Lake WA the major change ecosystem change is related to sewage in the lake – adding wastewater plants in the 1960s reduced phosphorous (Figure [S2.1](#)) and in turn, blue-green algae (Figure [S2.2](#)) which can be seen by examining time series plots of the data.

We can then use the MARSS package to fit a DLM. We'll filter the data to only use counts before 1974, and log transform densities. We also z-score the covariate. This is a good example of imperfect data – there's a couple of missing observations in the covariate, which we can't have in a DLM or GAM – so let's just fill them in with a spline. We can do this as follows:

```
dat <- dplyr::filter(dat, Year < 1974) %>%  
  dplyr::mutate(y = log(Bluegreens), zp = as.numeric(scale(TP)))  
  
interpolated <- spline(dat$zp, xout = 1:nrow(dat))$y  
dat$zp[which(is.na(dat$zp))] <- interpolated[which(is.na(dat$zp))]
```

The response data exhibit a strong seasonal cycle – and it's a good idea to de-season that to focus on the phosphorous effect. Here we de-season the data:

```
# Fill in a few missing values  
dat$y_interp <- dat$y  
interpolated <- spline(dat$y, xout = 1:nrow(dat))$y  
missing <- which(is.na(dat$y_interp))  
dat$y_interp[missing] <- interpolated[missing]
```

```
dat <- dplyr::group_by(dat, Month) %>%
  dplyr::mutate(month_mean = mean(y, na.rm=T)) %>%
  dplyr::ungroup() %>%
  dplyr::mutate(y_adj = y_interp - month_mean)
```

Last, we should look at the lags between driver and response relationships which we can do using a CCF (Figure S2.3). The CCF here indicates a 2-4 month lag, where TP and algae are more strongly associated with one another.

We'll use a lag of 4 months, which we can apply as follows:

```
lag_n <- 5
dat$lagged_zp <- c(rep(NA, lag_n), dat$zp[1:(nrow(dat)-lag_n)])
dat <- dat[-c(1:lag_n),]
```

DLMs

First, we need to define the model – this block is basically copied from the MARSS book, [the salmon survival case study](#)

```
m <- 2
TT <- nrow(dat)
B <- diag(m) ## 2x2; Identity
U <- matrix(0, nrow = m, ncol = 1) ## 2x1; both elements = 0
Q <- matrix(list(0), m, m) ## 2x2; all 0 for now
diag(Q)[1] <- 0.0000001 ## time-independent intercept
diag(Q)[2] <- c("q.beta") ## time-dependent slope
#diag(Q) <- c("q.alpha", "q.beta") ## 2x2; diag = (q1,q2)
Z <- array(NA, c(1, m, TT)) ## NxMxT; empty for now
Z[1, 1, ] <- rep(1, TT) ## Nx1; 1's for intercept
Z[1, 2, ] <- dat$lagged_zp ## Nx1; predictor variable
A <- matrix(0) ## 1x1; scalar = 0
R <- matrix("r") ## 1x1; scalar = r
x0 = matrix(c(0, 0)) ## initial conditions known
## only need starting values for regr parameters
inits_list <- list(x0 = matrix(nrow = m))
## list of model matrices & vectors
mod_list <- list(B = B, U = U, x0 = x0, Q = Q, Z = Z, A = A, R = R)
## convert response to matrix
dat_mat <- matrix(dat$y_adj, nrow = 1)
## fit the model -- crank up the maxit to ensure convergence
```

```
d1m_1 <- MARSS(dat_mat, inits = inits_list, model = mod_list,
control = list(maxit=4000))
```

Success! abstol and log-log tests passed at 19 iterations.
Alert: conv.test.slope.tol is 0.5.
Test with smaller values (<0.1) to ensure convergence.

MARSS fit is
Estimation method: kem
Convergence test: conv.test.slope.tol = 0.5, abstol = 0.001
Estimation converged in 19 iterations.
Log-likelihood: -168.2105
AIC: 340.4209 AICc: 340.5091

	Estimate
R.r	0.311
Q.q.beta	0.227

Initial states (x0) defined at t=0

Standard errors have not been calculated.
Use MARSSparamCIs to compute CIs and bias estimates.

Let's put the state estimates back into the original dataframe and do some plotting. The slope is generally positive, which is what we expect for the hypothesized relationship between bluegreen algae and total phosphorus – and exhibits clear variation through time (Figure S2.4). We can also plot the fitted values from our model (Figure S2.5).

And we can examine the residuals for a temporal trend, and see that because the DLM is flexible, there's not much evidence of a trend (Figure S2.6) and there are no issues with the q-q plot (Figure S2.7).

GAMs

We can fit a GAM model 5 ways, 1) so that the smoother is a function of phosphorus which represents a nonlinear functional relationship between total phosphorus and bluegreen algae that is not temporally structured thus the relationship is stationary and fixed through time. This is how GAMs are typically structured for ecological analyses. 2) a temporally structured model such that the abundance of bluegreen algae varies as a function of time unrelated to phosphorus, 3) A temporally structured model so that the smooth function of phosphorus varies by time, and 4) a smooth on date as a driver of bluegreen algae where the smooth through time varies by phosphorus level. Finally, these relationships can be fit as 5) a tensor product

which provide a multivariate function spanning the dimensions of time and phosphorus. We will not discuss tensor products in greater detail and only include them for completeness. We show the model predicted bluegreen algae trend through time using these five modeling approaches (Figure S2.8). While they are quite similar, the predictions diverge substantially at the end of the time series.

```
dat$n_date <- as.numeric(as.factor(dat$date))

gam_cr1 <- gam(y_adj ~ s(lagged_zp, bs = "cr"), data = dat) #1
gam_cr2 <- gam(y_adj ~ s(lagged_zp, by = n_date, bs = "cr"), data = dat) #2
gam_cr3 <- gam(y_adj ~ s(n_date, by = lagged_zp, bs = "cr"), data = dat) #3
gam_cr4 <- gam(y_adj ~ s(n_date), data = dat) #4
gam_cr5 <- gam(y_adj ~ te(lagged_zp, n_date), data = dat) #5
```

It is important to also examine the residuals from these models. For example, we can look at the residuals for the GAMs fit to the Lake Washington data. In looking across models, those that do not include time-dependent parameters show evidence of a persistent trend in the residuals that is not accounted for by the model (Figure S2.9) despite the q-q plot being reasonable for all model specifications (Figure S2.10).

```
resid_df_1 <- data.frame(date = dat$date, n_date=dat$n_date, resid = resid(gam_cr1),
                        Model = "s(Phos)")
resid_df_2 <- data.frame(date = dat$date, n_date=dat$n_date, resid = resid(gam_cr2),
                        Model = "s(Phos, by = date)")
resid_df_3 <- data.frame(date = dat$date, n_date=dat$n_date, resid = resid(gam_cr3),
                        Model = "s(date, by=Phos)")
resid_df_4 <- data.frame(date = dat$date, n_date=dat$n_date, resid = resid(gam_cr4),
                        Model = "s(date)")
resid_df_5 <- data.frame(date = dat$date, n_date=dat$n_date, resid = resid(gam_cr5),
                        Model = "te(Phos, date)")
resid <- rbind(resid_df_1, resid_df_2, resid_df_3, resid_df_4, resid_df_5)%>%
  left_join(p_values)

gam_resid<-ggplot(resid, aes(date, resid, group = Model)) +
  geom_point() +
  geom_smooth(method = "gam", formula = y ~ s(x)) + theme_classic() +
  geom_text(
    aes(x = as.Date('1965-11-01'), y = -3, label =paste0("p = ",round(p_value,4))),
    inherit.aes = FALSE)+
  facet_wrap(~ Model) +
  xlab("Date") +
  ylab("Residuals") +
  ggtitle("GAM Residuals")
```

There are a few more analysis approaches for running GAMs including the choice of smoother function (Figure S2.8 & Figure S2.11) and the number of knots, or the “wiggleness” (Figure S2.12) of the smoothed relationship. We can plot these different results and include the results of the DLM for comparison, note that ‘cr’ and ‘tp’ are so similar they are difficult to discern from the plot. We can do a similar comparison for the choice of k. Here we compare a four different numbers of knots (Figure S2.12), we show an intentionally wide range of k (30 - 120) to illustrate the effects of k on estimated uncertainty, in practice k should be specified based on degrees of freedom and expectations of variability through time.

LM

Pollution reduction occurred over a multi-year period (1964 - 1968) and as such the change in the relationship between phosphorus and cyanobacteria is likely a continuous process evolving over time. Here we demonstrate a LM approach for the Lake Washington, choosing a year in the middle of the time period as the temporal break (1966). We find these results agree with our GAM/DLM results such that the best model (Table S2.1) includes a time-dependent slope where prior to 1966 there is a strong positive relationship between total phosphorus and cyanobacteria prior to 1966 and this relationship is reduced to nearly 0 after 1966 (Table S2.2). Given expected change in the relationship is supposed to be gradual through time, the hypothesized mechanism for the change in the relationship supports using a GAM or DLM rather than an LM but we include the LM to show agreement across the three model types for completeness.

Figures & Tables

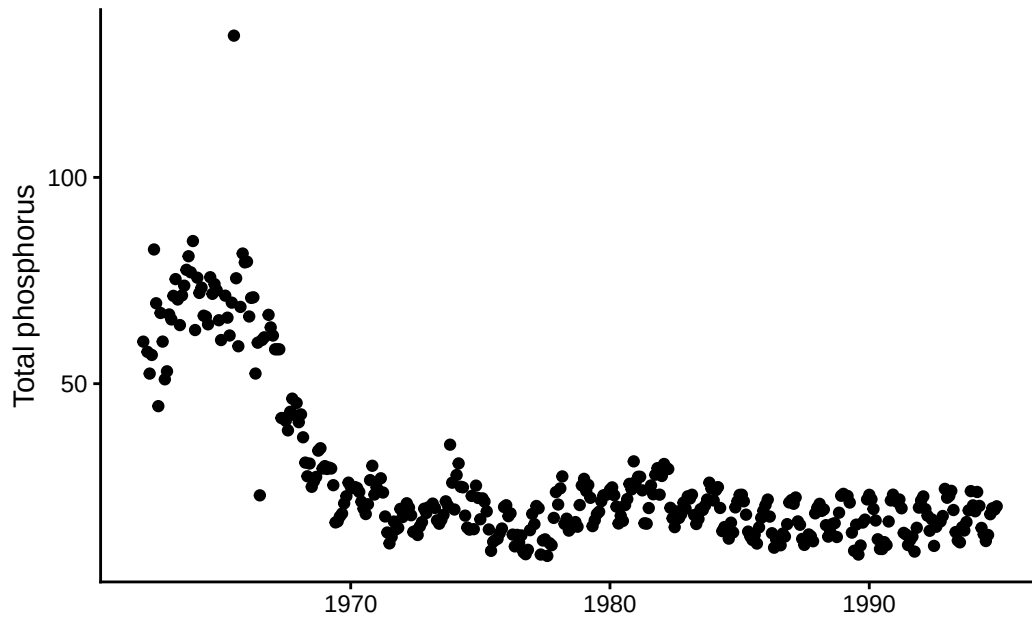


Figure S2.1: Time series of total phosphorus in Lake Washington



Figure S2.2: Time series of bluegreen algae (cyanobacteria) abundance in Lake Washington

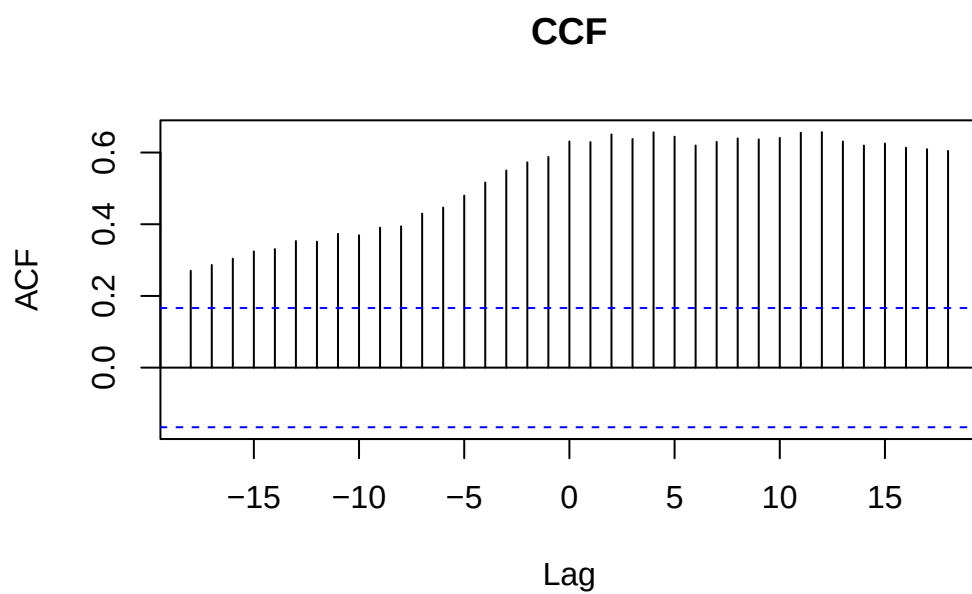


Figure S2.3: A CCF (Cross-Correlation Function) plot that represents the correlation between total phosphorus and bluegreen algae abundance as a function of the lag (time difference) between them.

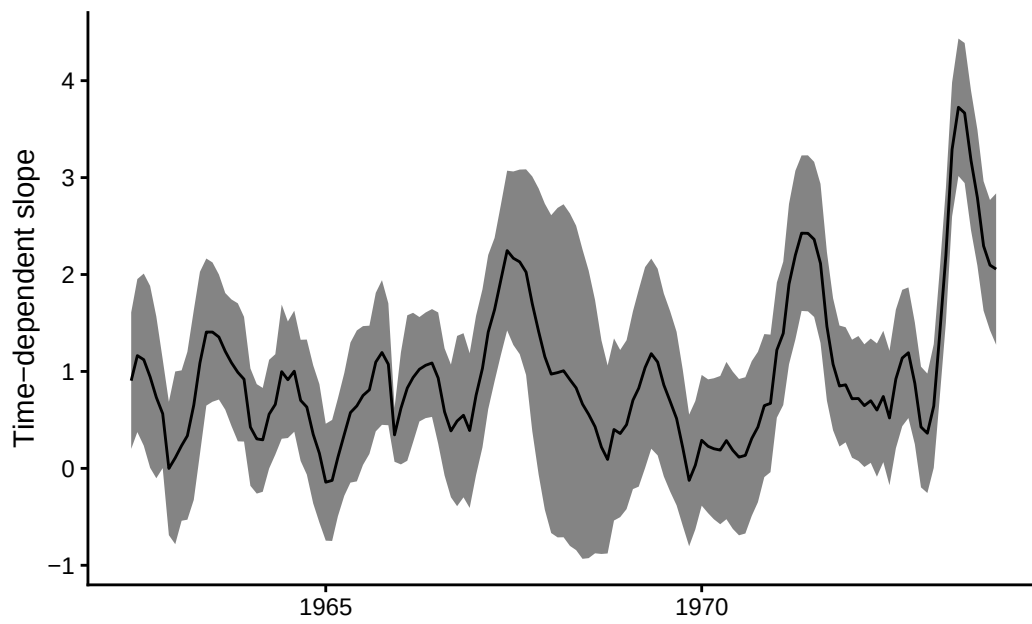


Figure S2.4: Time-dependent slope of the relationship between bluegreen algae (response) and total phosphorus (driver) from a DLM

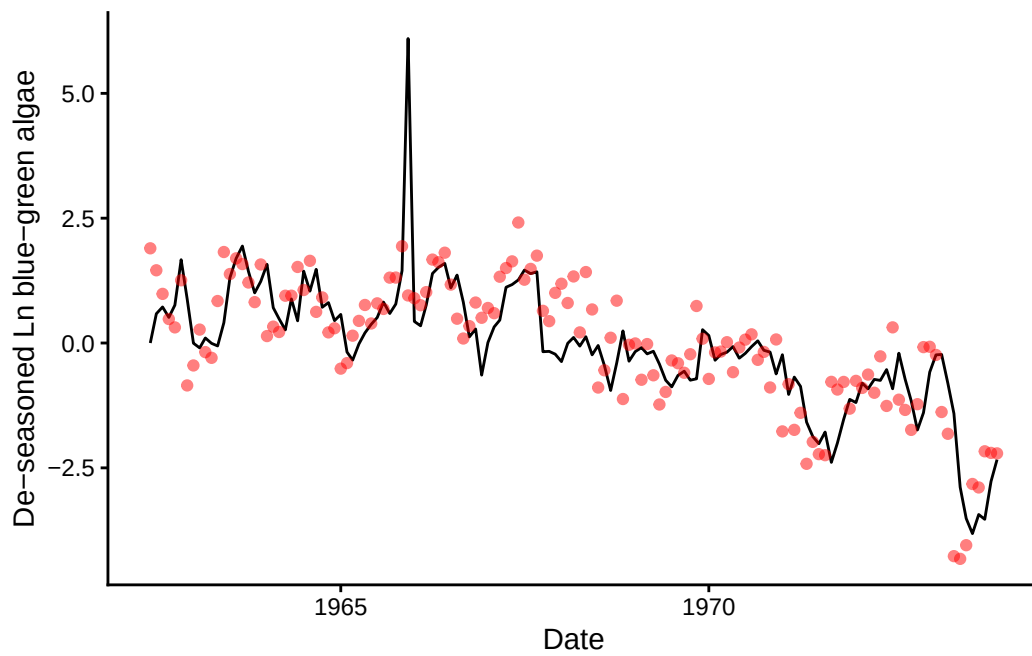


Figure S2.5: Fit of the DLM model with time-dependent slope (black line) to observed blue-green algae abundance (red points).

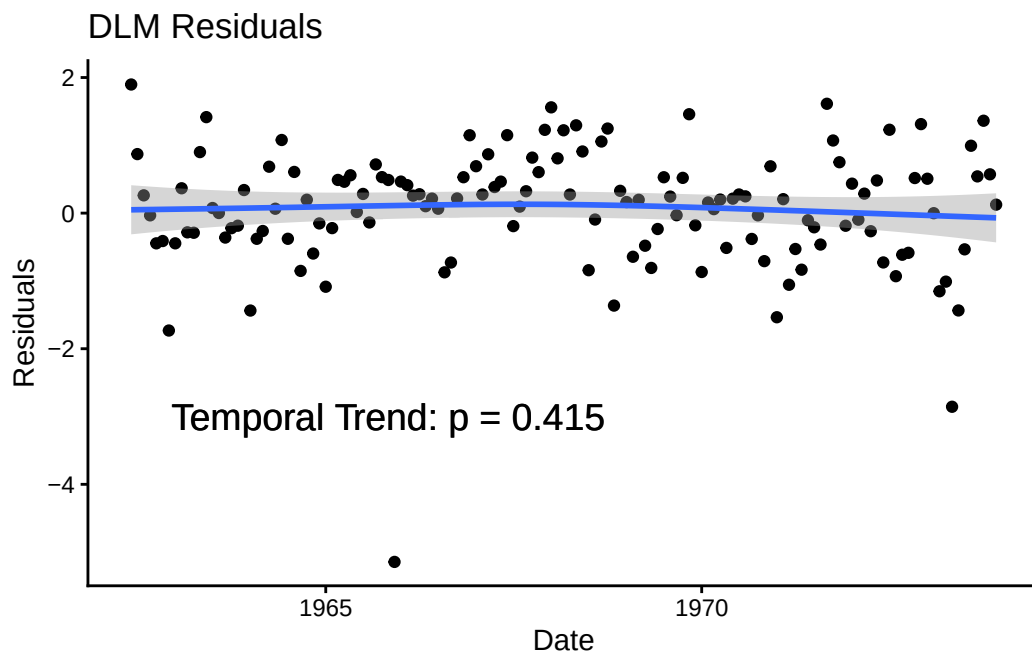


Figure S2.6: Time series of residuals from the DLM model fit with a time-dependent slope between blue-green algae and phosphorus. p-value denotes the significance of the trend in residuals through time.

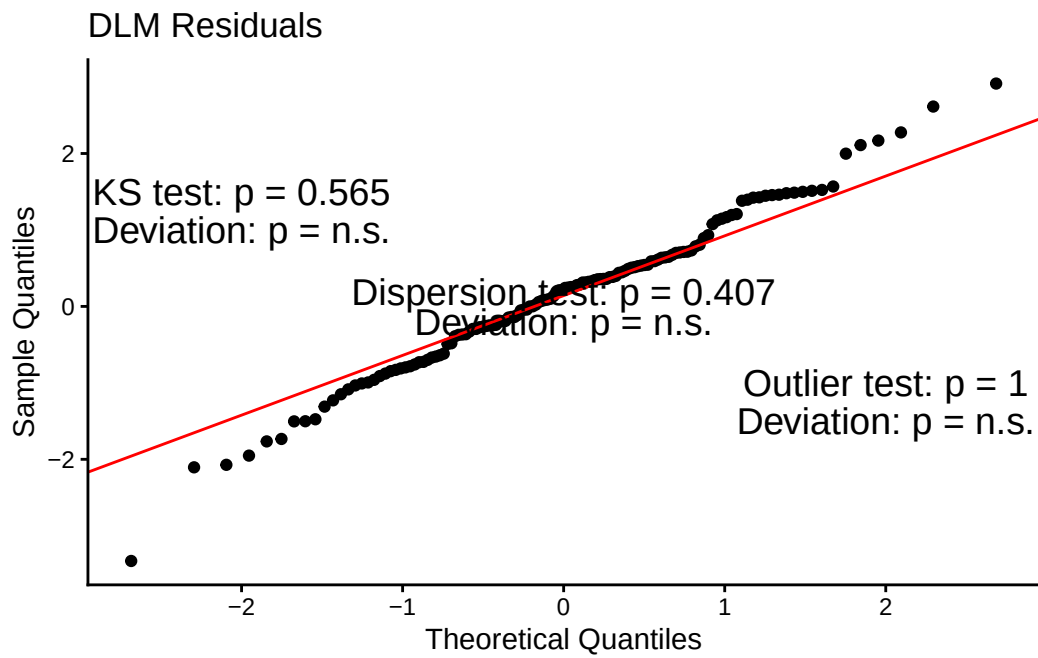


Figure S2.7: Q-Q plot from the DLM model fit with a time-dependent slope between blue-green algae and phosphorus

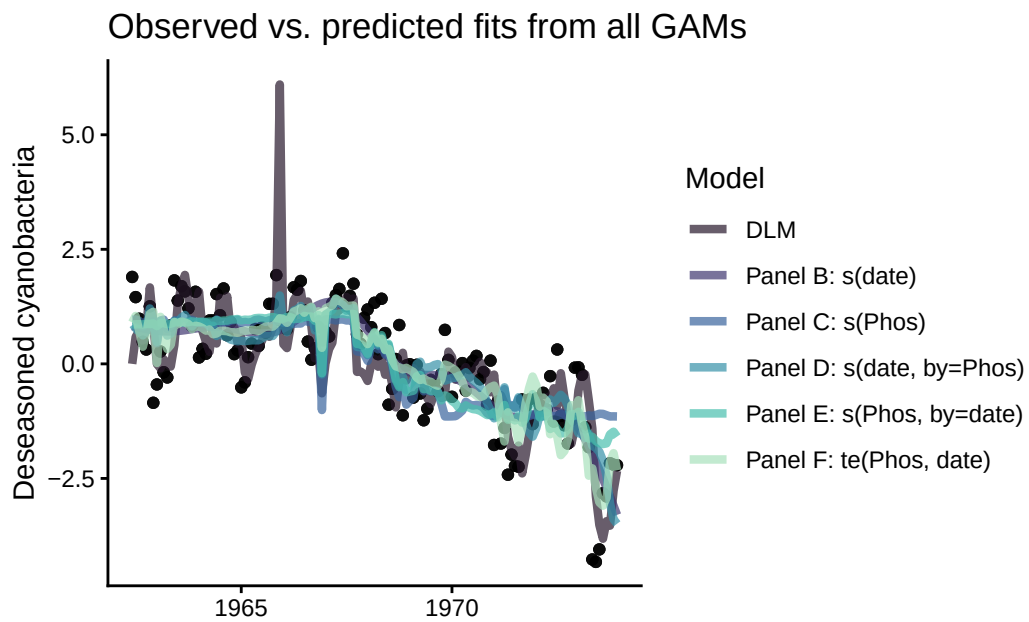


Figure S2.8: Fit of the GAM models (lines) to observed bluegreen algae (cyanobacteria) abundance (black points). Colors denote different specifications of the smoother functions.

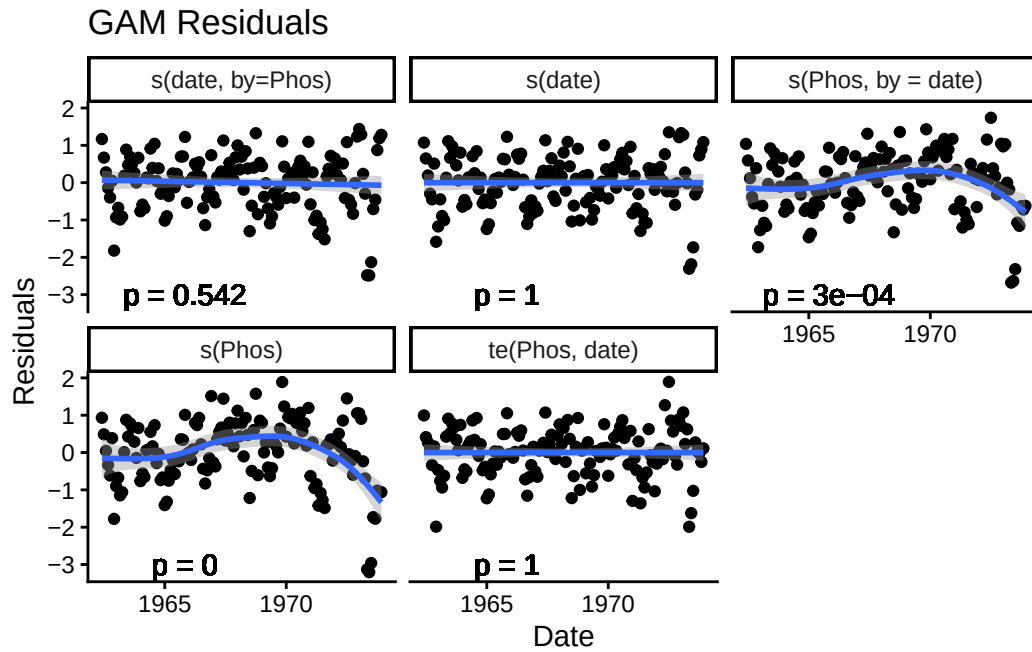


Figure S2.9: Time series of residuals from the five GAM models where residuals from models that do not have a temporally structured smooth term ($s(\text{Phos})$ and $s(\text{Phos}, \text{by} = \text{date})$) have a trend through time. p-value denotes the significance of the trend in residuals through time.

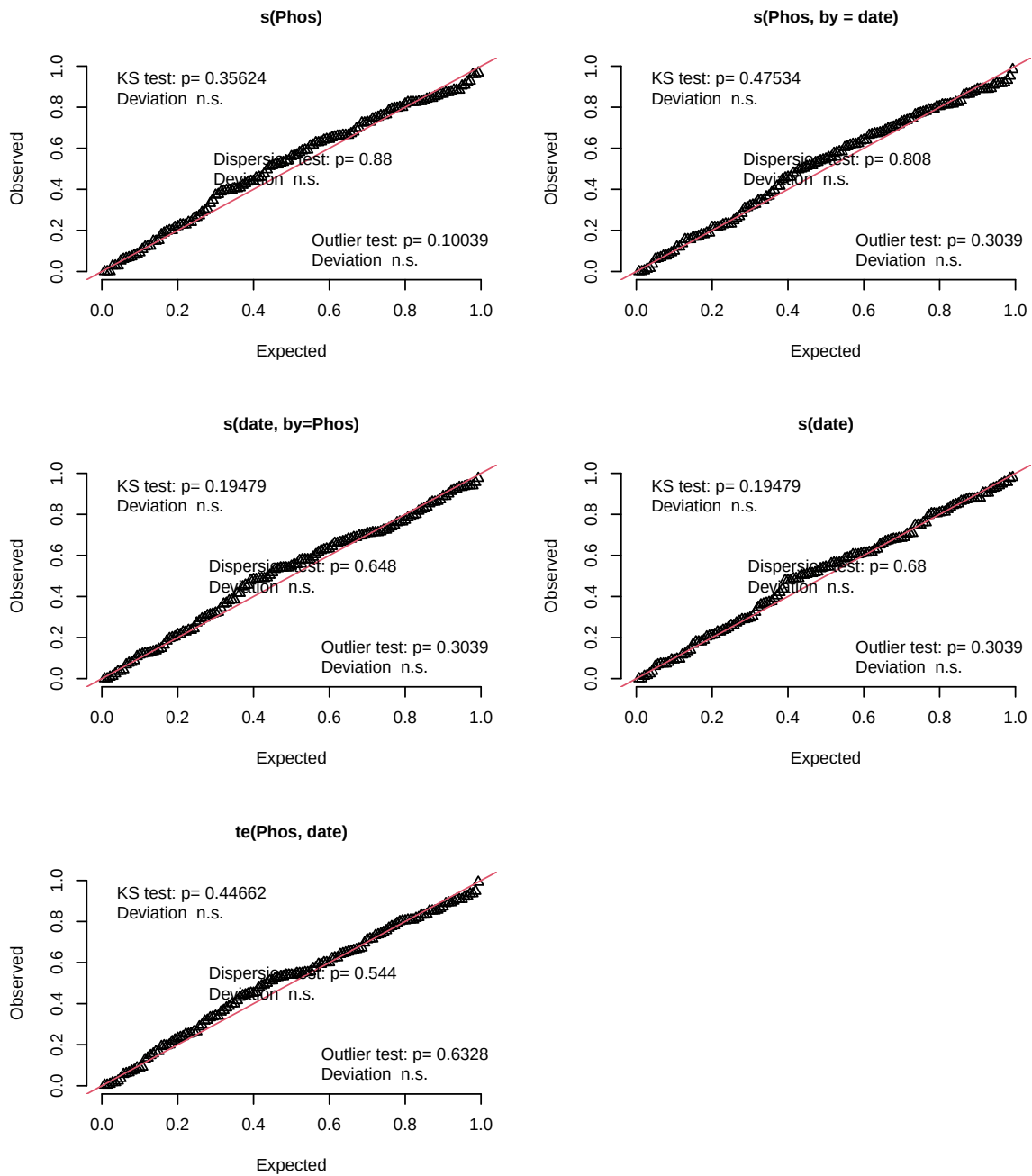


Figure S2.10: Q-Q for each of the five GAM models where residuals from models that do not have a temporally structured smooth term (**s(Phos)** and **s(Phos, by = date)**) have a trend through time but no significant issues with overdispersion, outliers, or deviations from the 1:1 line.

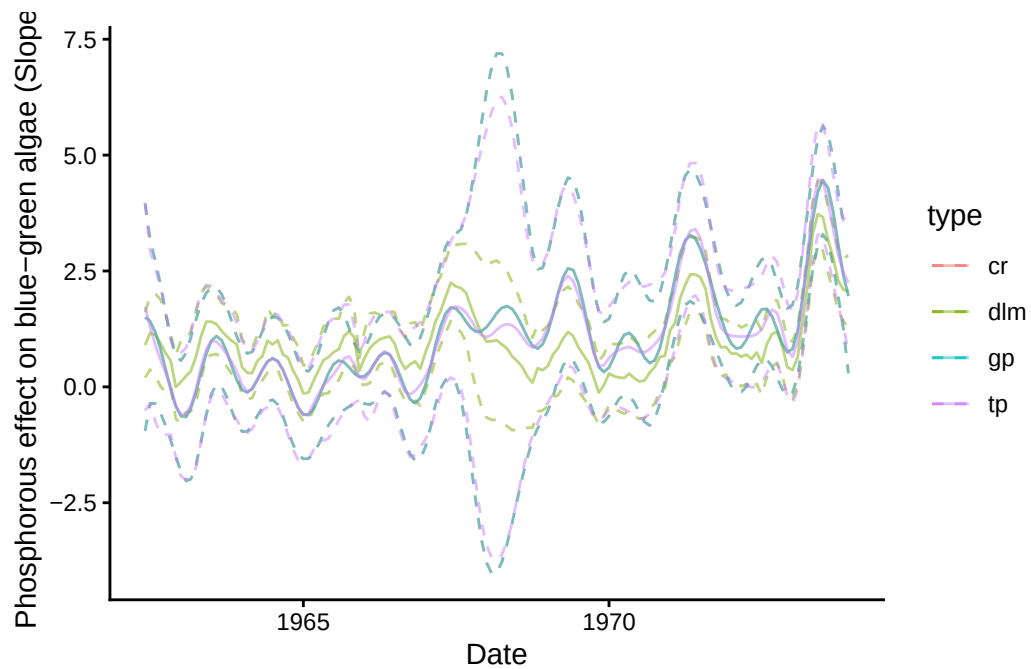


Figure S2.11: Effect of the choice of smoother function on estimates of time-dependent slope parameter where color denotes model type. 3 GAM smoother types (cr, tp, gp) are contrasted with a DLM (dlm).

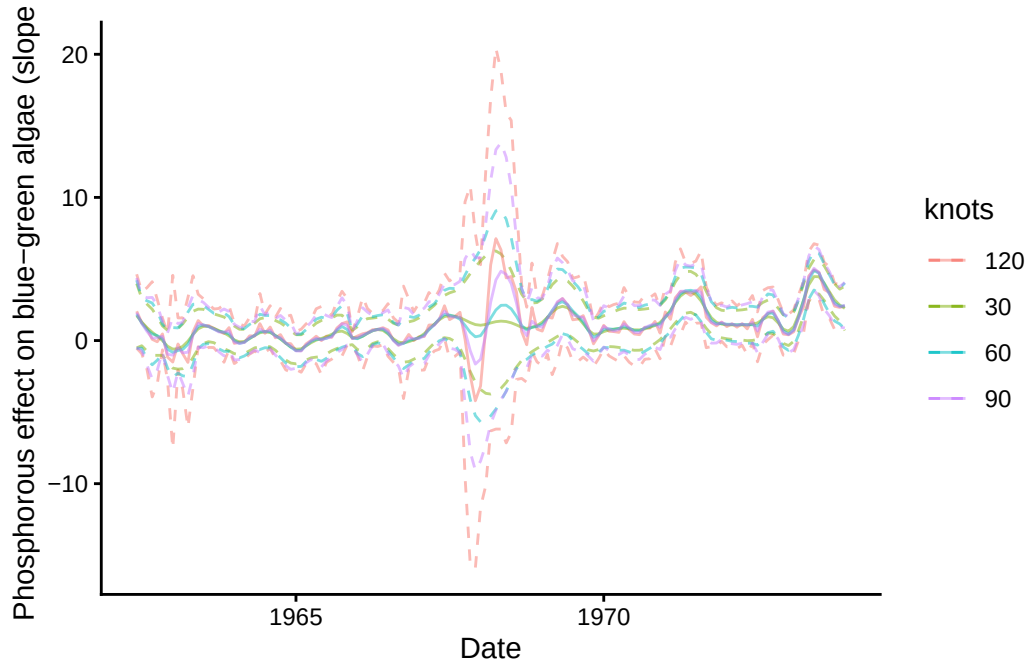


Figure S2.12: Effect of smoothing degrees of freedom on the standard errors of the estimates of time-dependent slope parameter where color denotes model type. We show 4 distinct knots (colors)

Model	AIC
Time-Independent Intercept	469.15
Time-Dependent Intercept	388.96
Time-Independent Slope	430.12
Time-Dependent Slope	372.37

Table S2.1: AIC values comparing linear models with time-dependent abundance models versus nonstationary relationship models

term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.35	0.14	2.47	0.01
lagged_zp	1.45	0.17	8.62	0.00
as.factor(period)Treatment	0.21	0.29	0.71	0.48
lagged_zp:as.factor(period)Treatment	-1.22	0.27	-4.52	0.00

Table S2.2: Model coefficients for the nonstationary relationship model between total phosphorus and cyanobacteria

Appendix 3: Gulf of Alaska salmon analysis

Appendix 3: Gulf of Alaska salmon analysis

Here we show the LM, GAM, and DLM analysis for the Gulf of Alaska salmon dataset. For complete code, see the [full quarto document](#)

Data Processing

First, we standardize and examine the trends in SST and salmon abundance from Litzow et al. 2018 updated with data through 2025 (Figure [S3.1](#) & Figure [S3.2](#)).

LM

We start our investigation using linear models, to evaluate the relationship between salmon catch and SST anomalies with periods defined *a-priori* as before and after 1988/1989 (from Litzow et al. 2018). Previous investigations used this approach, and while we primarily discuss DLMs and GAMs for this dataset in the main text, we include this illustrative example here for completeness and comparability to the previous study.

We compare linear model structures using classic model selection criteria, AIC. First, we examine a 1) intercept only model, 2) salmon abundance with period as a predictor, 3) salmon abundance with period and SST anomalies as predictors, and 4) an interaction between SST anomalies and period. This shows that model 2 is a better fit to the data than model 1 indicating that a time-dependent intercept is an improvement to model fit than a time-independent intercept. Similarly, model 4 is the best model showing that a time-dependent slope with SST is an improvement to a time-independent slope (Table [S3.1](#)).

Functionally, model 4 has a time-dependent slope and a time-dependent intercept which we can see by both plotting the relationships and examining the model output. We can see a strong positive correlation prior to 1988 (Figure [S3.3](#)) followed by a weak negative relationship which is represented in the slopes of the relationship (Table [S3.2](#) & Figure [S3.4](#)).

GAM

Next, we consider generalized additive models. We fit two distinct GAM models to the salmon catch data and SST anomalies to illustrate the differences in a model with a time-dependent slope and a time-dependent intercept. We find that when fitting a time-dependent slope and time-dependent intercept model, most of the variability in the relationship is explained by the time-dependent intercept (Figure S3.5) while the time-dependent slope does not change much through time (Figure S3.6). When only a time-dependent slope is fit to the data, the slope parameter is much more variable through time (Figure S3.6), and looks similar to the slope estimated by the linear models (Figure S3.4).

Next, we check the residuals plots for both parameterizations of the models. The time-dependent slope model appears to have some residual temporally variability while there are no trends through time in the model with both a time-dependent slope and a time-dependent intercept (Figure S3.7) and the qq-pot looks reasonable for both (Figure S3.8).

We need to decide if there is enough ecological evidence to support a model with both a time-dependent slope and time-dependent intercept and whether that model has signs of overfitting. We can do this by using cross validation and comparing the root mean square error (RMSE) of a set of training data compared to test data. For time series data its important that cross validation methods retain the temporal structure of the data. To do this, we used approximate leave-future out cross validation (LFO-CV) where we fit the model to 15 years of data (training data, first year to year $n-1$) and predict 1 year ahead (test data, year n). The model is used to produce in-sample predictions for the training data and one-year-ahead out of sample predictions for the test data. Then the model is iteratively refit walking one year ahead such that the training data then includes n and the test data is $n+1$. RMSE is then calculated for both the training and the test datasets based on predictions for each iteration. If RMSE of the training data is significantly lower than RMSE of the test data, it indicates the model is likely overfitting.

We compared LFO-CV for the training and test data for both candidate models: time-dependent slope and intercept, and time-dependent slope only (Figure S3.9). For the time-dependent slope and intercept model, the RMSE of the training data was significantly lower ($p < 0.05$) than the RMSE of the test data (Table S3.3). The RMSE of the training data was lower than the RMSE for the test data for the time-dependent slope model, but they were not significantly different ($p > 0.05$). Collectively with the residual diagnostics, we conclude that the time-dependent slope model is appropriate for interpreting ecological dynamics and performs better than the model that also included a time-dependent intercept (based on evidence of overfitting with this model parameterization). However, if the goal of fitting this model was prediction, it would be valuable to address the residual temporal variability that is not explained by SST and the time-dependent slope. Additional predictors and considering additional extrinsic factors that may contribute to the time-dependent salmon catch - SST relationship would be the next step.

DLM

We fit two distinct DLM models to the salmon catch data and SST anomalies to compare to the results of the GAM models. Generally, the parameters estimated by the DLM are more certain and follow similar patterns through time as the GAM (Figure [S3.10](#) & Figure [S3.11](#)). There is also more interannual variability in the parameter estimates of the DLM compared to the continuously smoothed pattern of the GAM. We also find similar residual patterns for the DLM compared to the GAM (Figure [S3.12](#) & Figure [S3.13](#)).

Figures and Tables

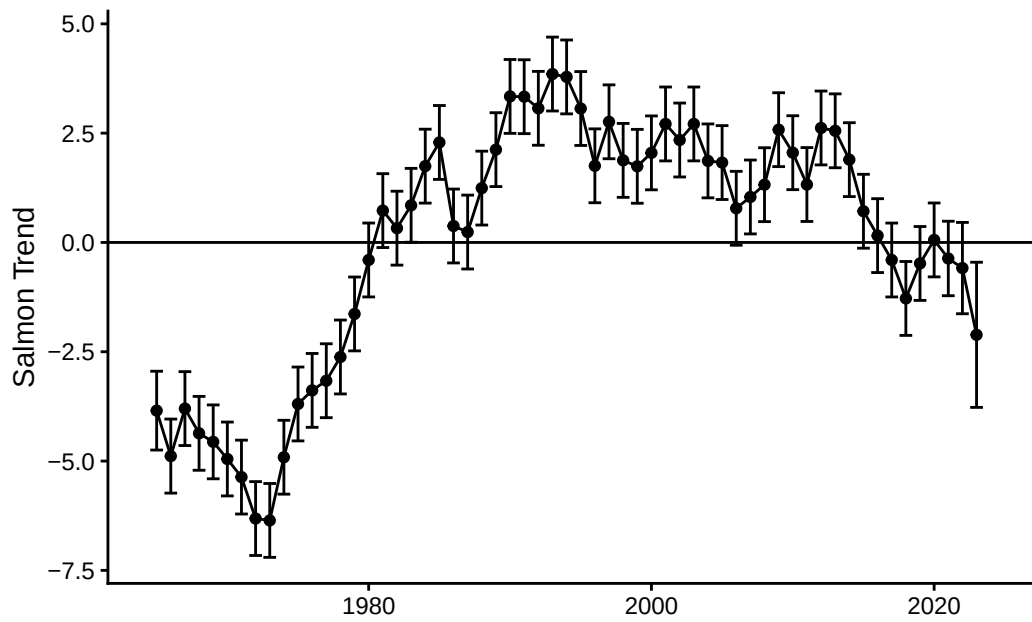


Figure S3.1: Time series of salmon catch from Lizow et al. 2018

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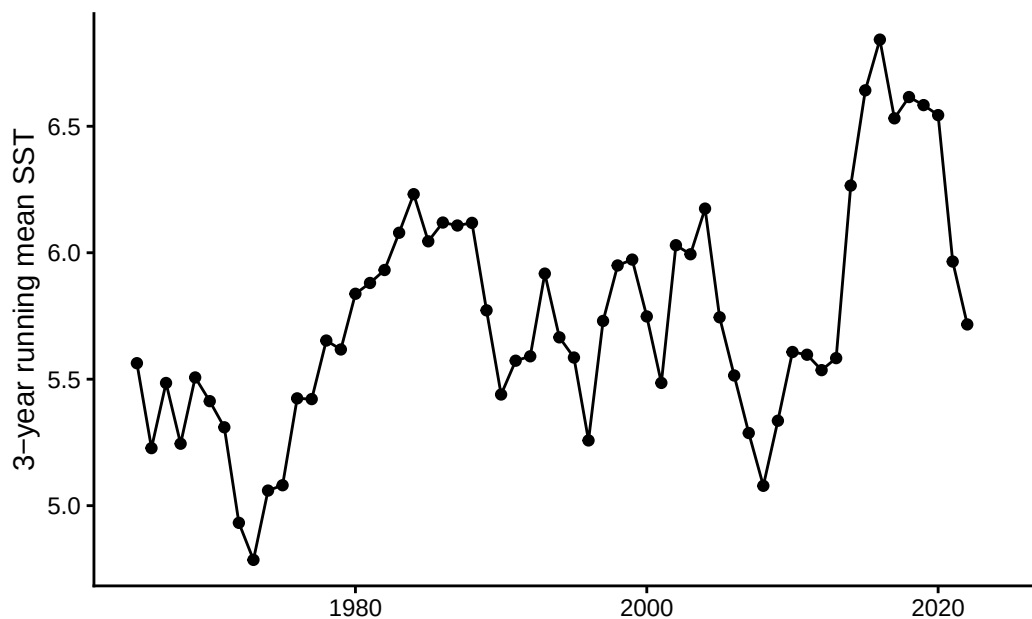


Figure S3.2: Time series of sea surface temperature anomalies from Lizow et al. 2018

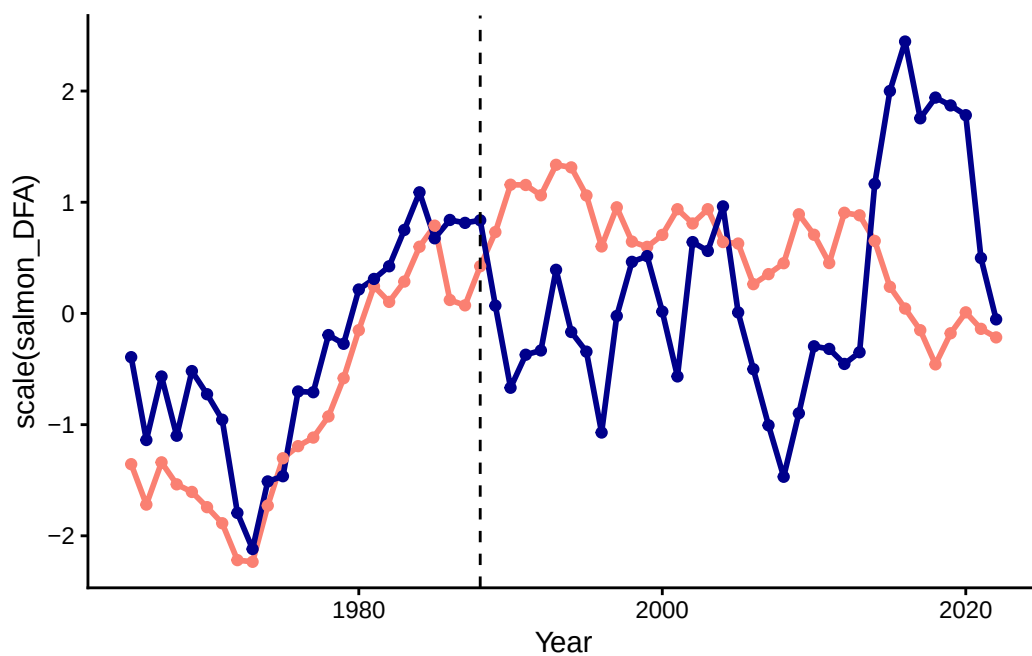


Figure S3.3: Time series of scaled salmon catch trend (salmon color) and SST anomalies (dark blue)

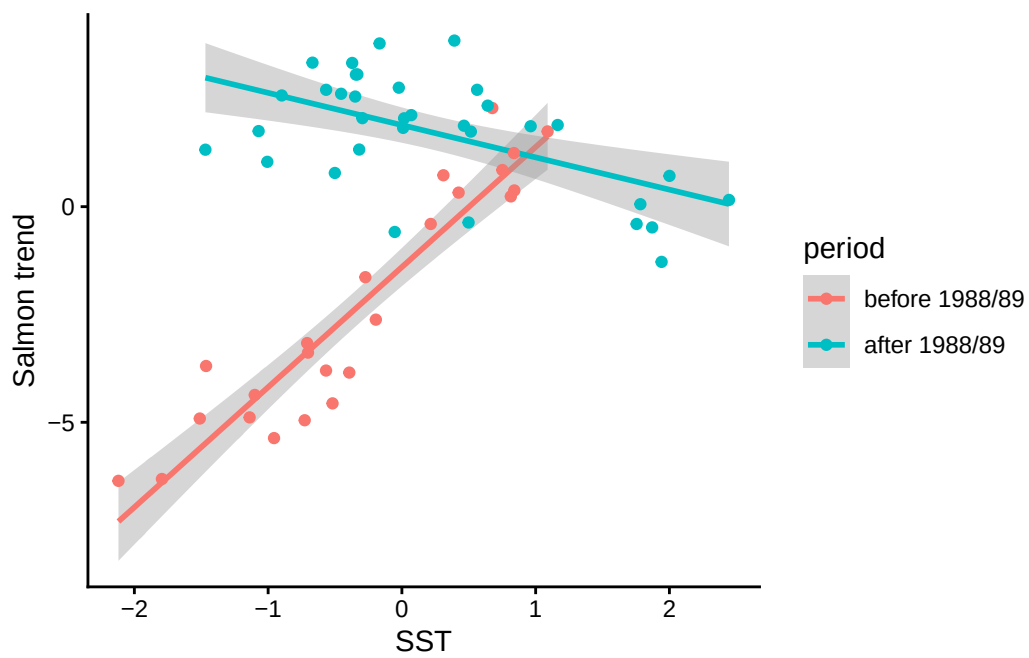


Figure S3.4: The relationship between salmon catch and SST anomalies before and after the 1988/1989 time periods

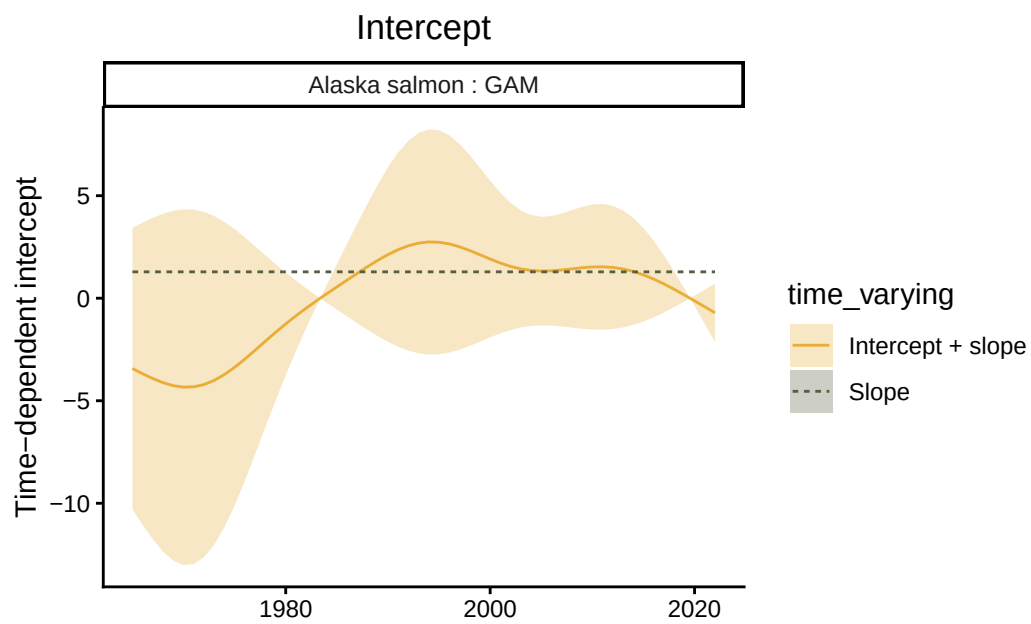


Figure S3.5: Time-dependent intercept of the GAM model representing the relationship between SST anomalies and salmon catch.

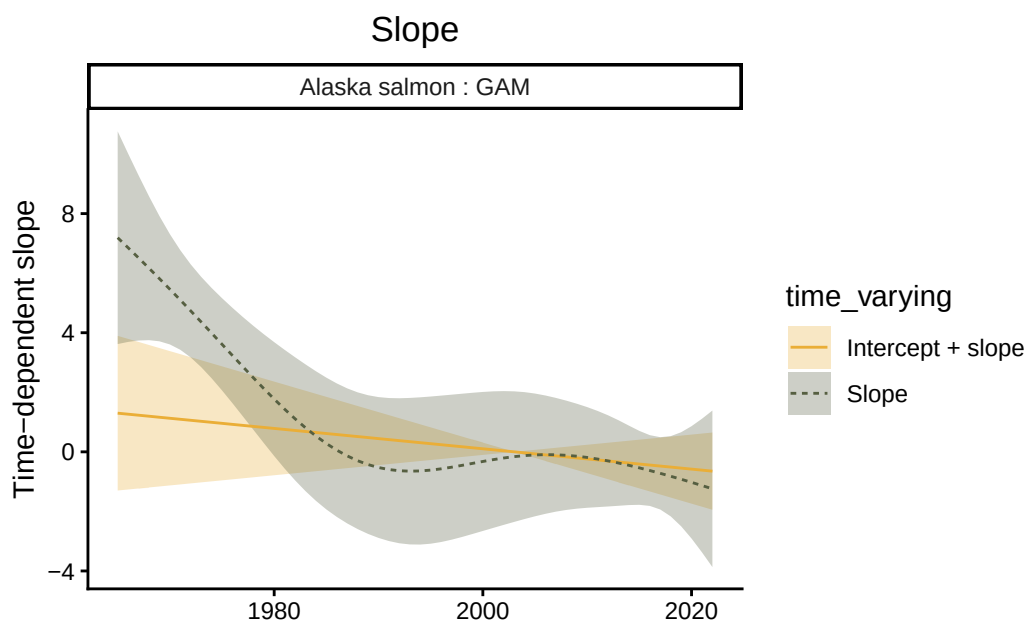


Figure S3.6: Time-dependent slope of the GAM model representing the relationship between SST anomalies and salmon catch.

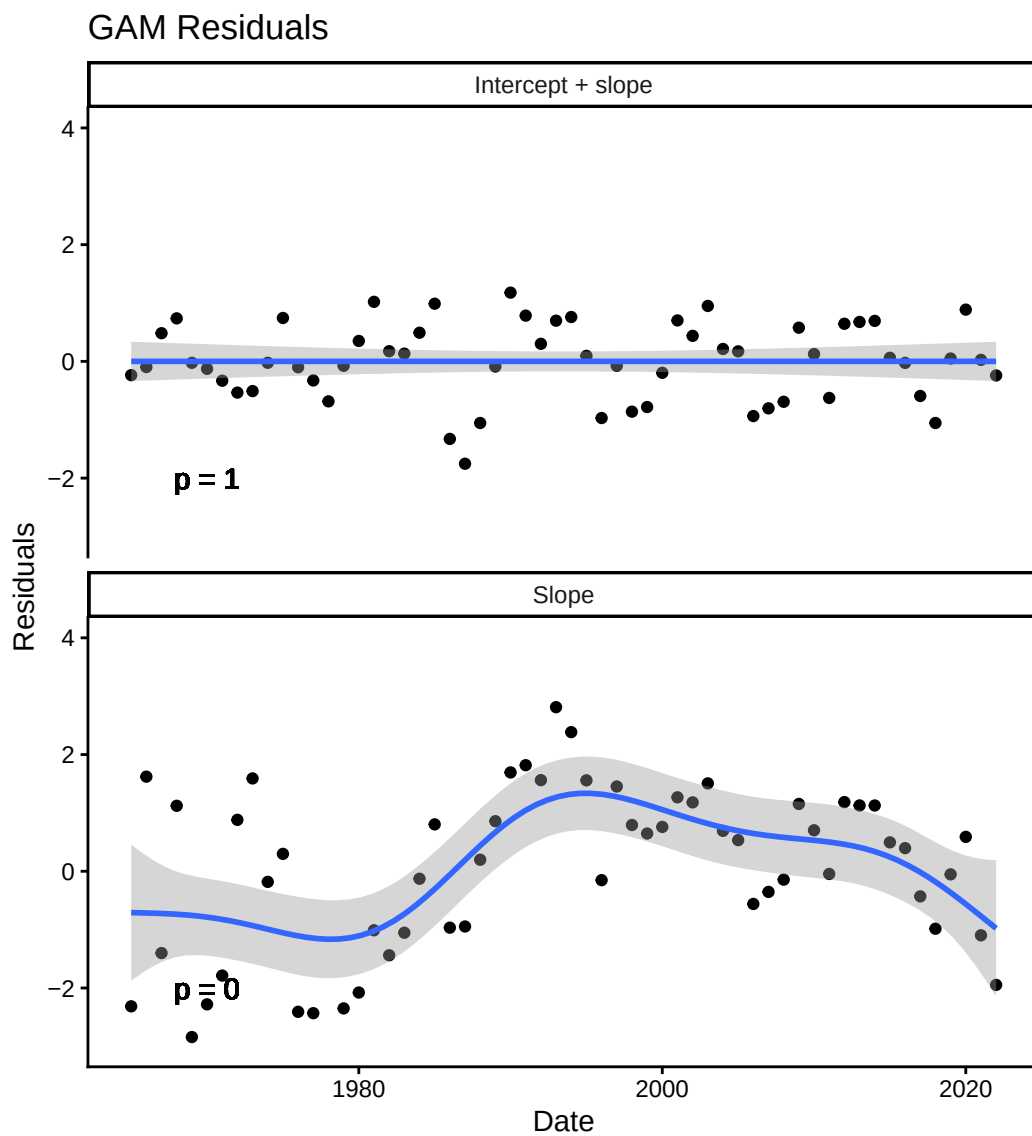


Figure S3.7: Time series of residuals from the GAM model fit with a time-dependent slope and both a time-dependent slope and intercept. p-value denotes the significance of the trend in residuals through time.

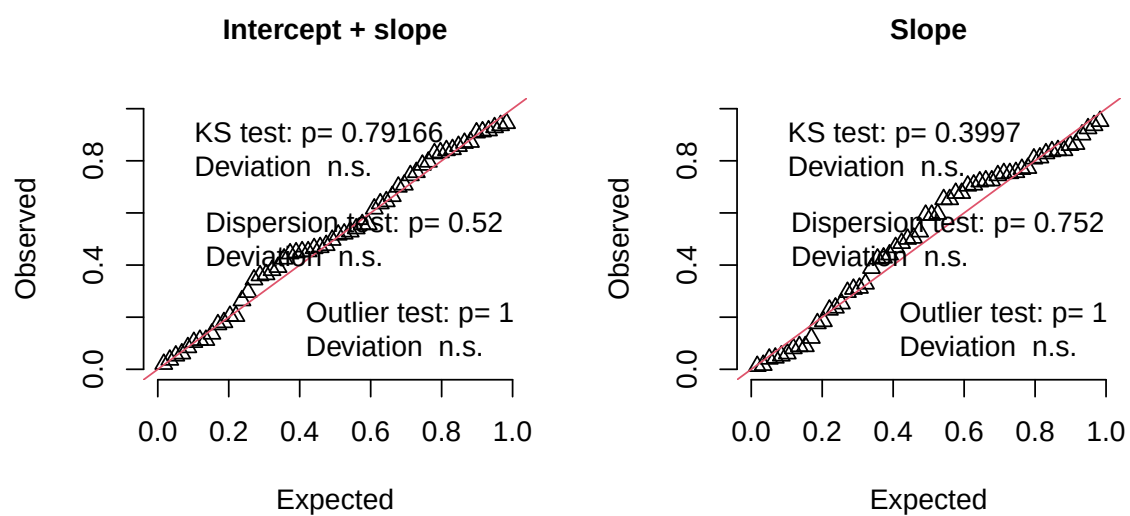
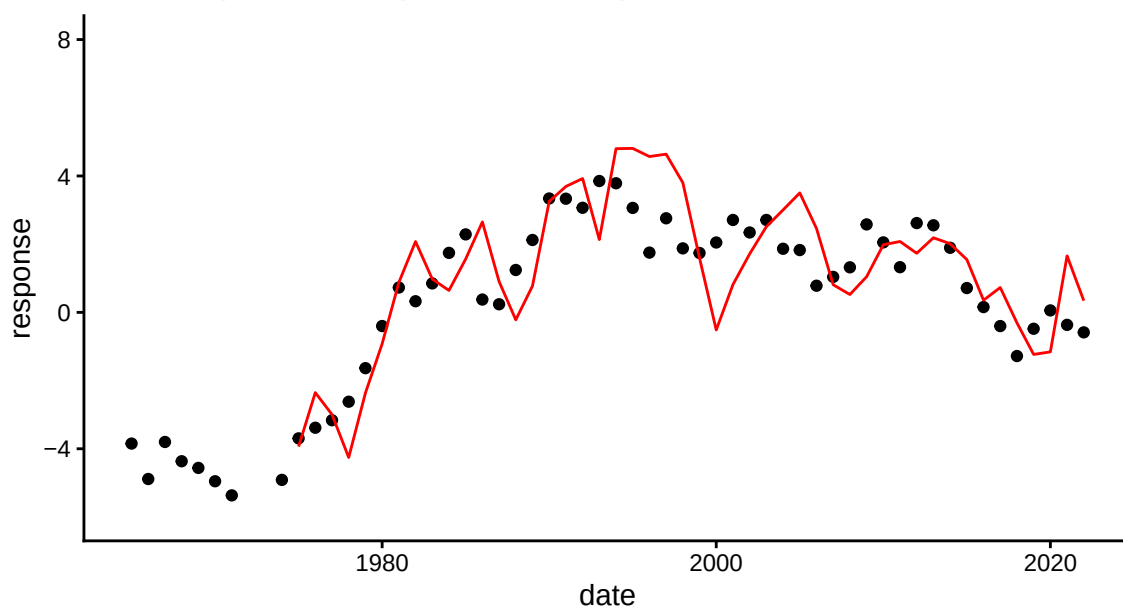


Figure S3.8: QQ plot from the GAM model fit with a time-dependent slope and both a time-dependent slope and intercept.

A. Time-dependent slope and intercept



B. Time-dependent slope

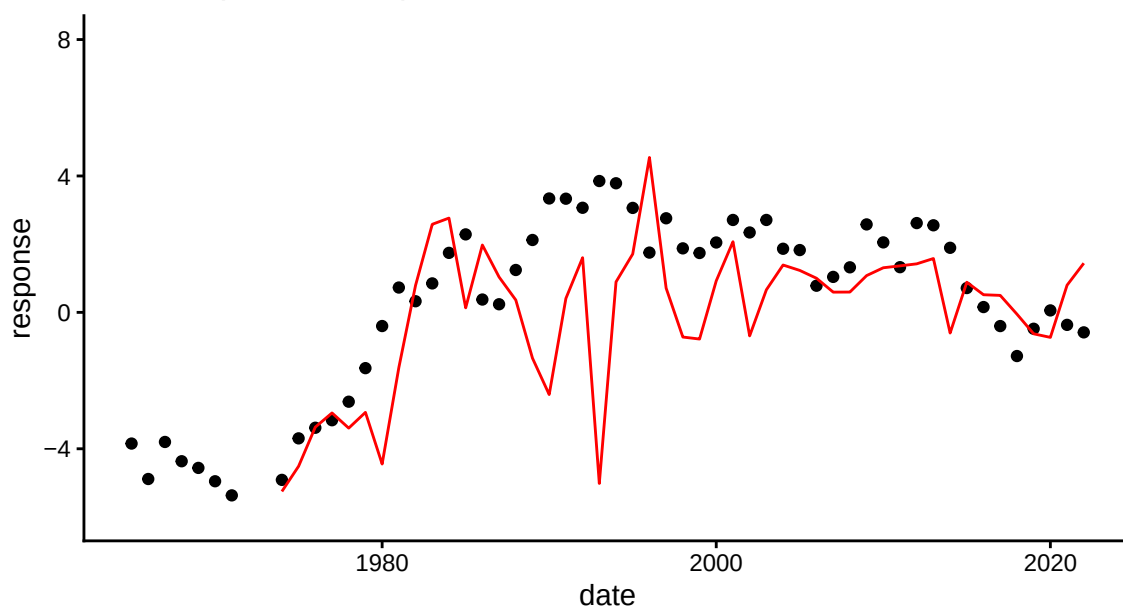


Figure S3.9: Leave-future-out cross validation for the GAM models fit with (A.) both a time-dependent slope and intercept, (B.) a time-dependent slope and and (C.).

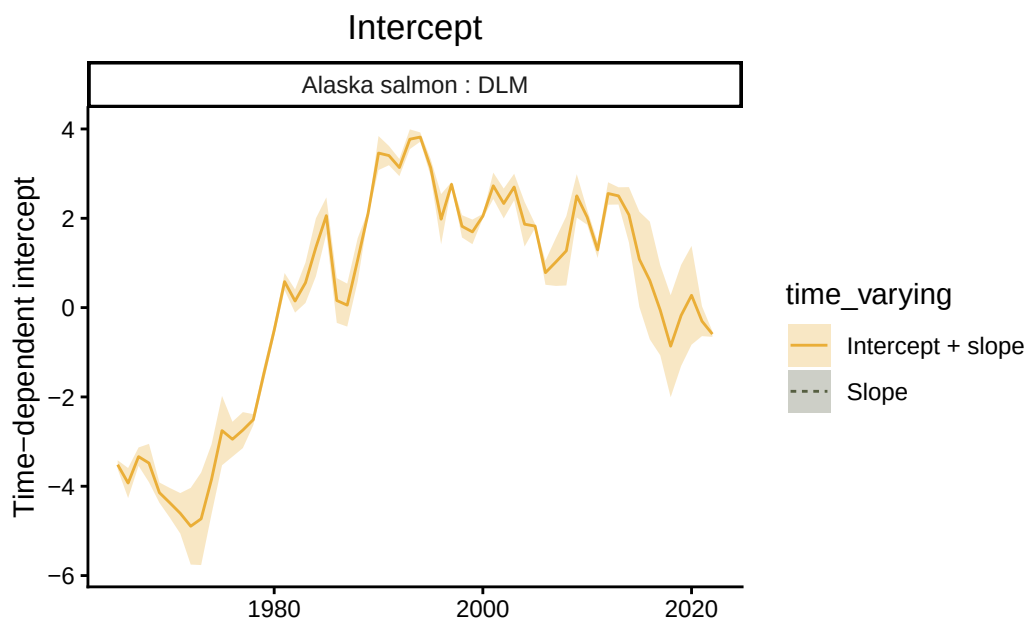


Figure S3.10: Time-dependent intercept of the DLM model representing the relationship between SST anomalies and salmon catch.

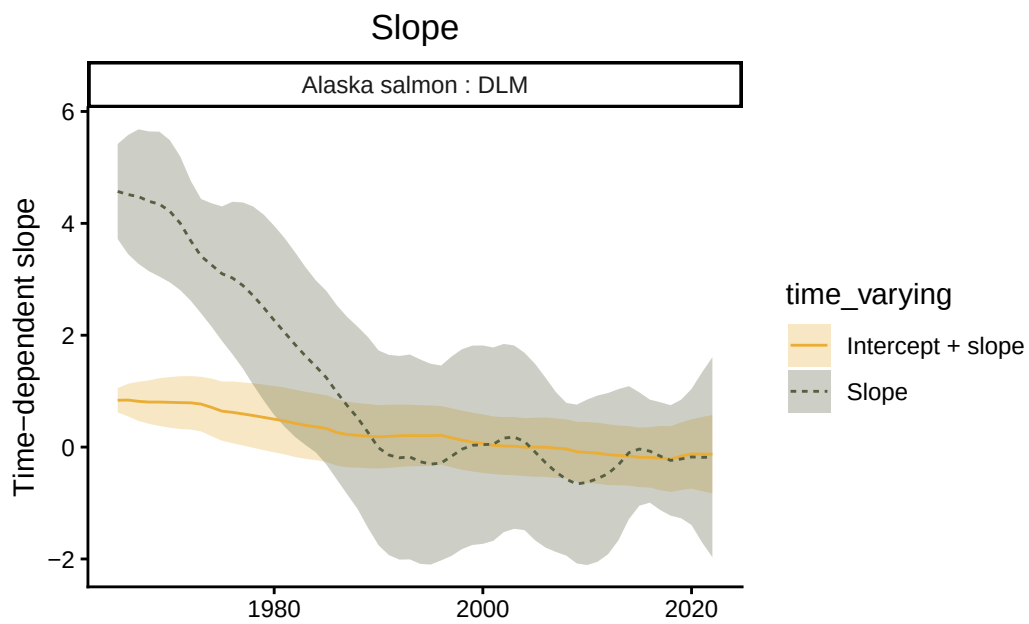


Figure S3.11: Time-dependent slope of the DLM model representing the relationship between SST anomalies and salmon catch.

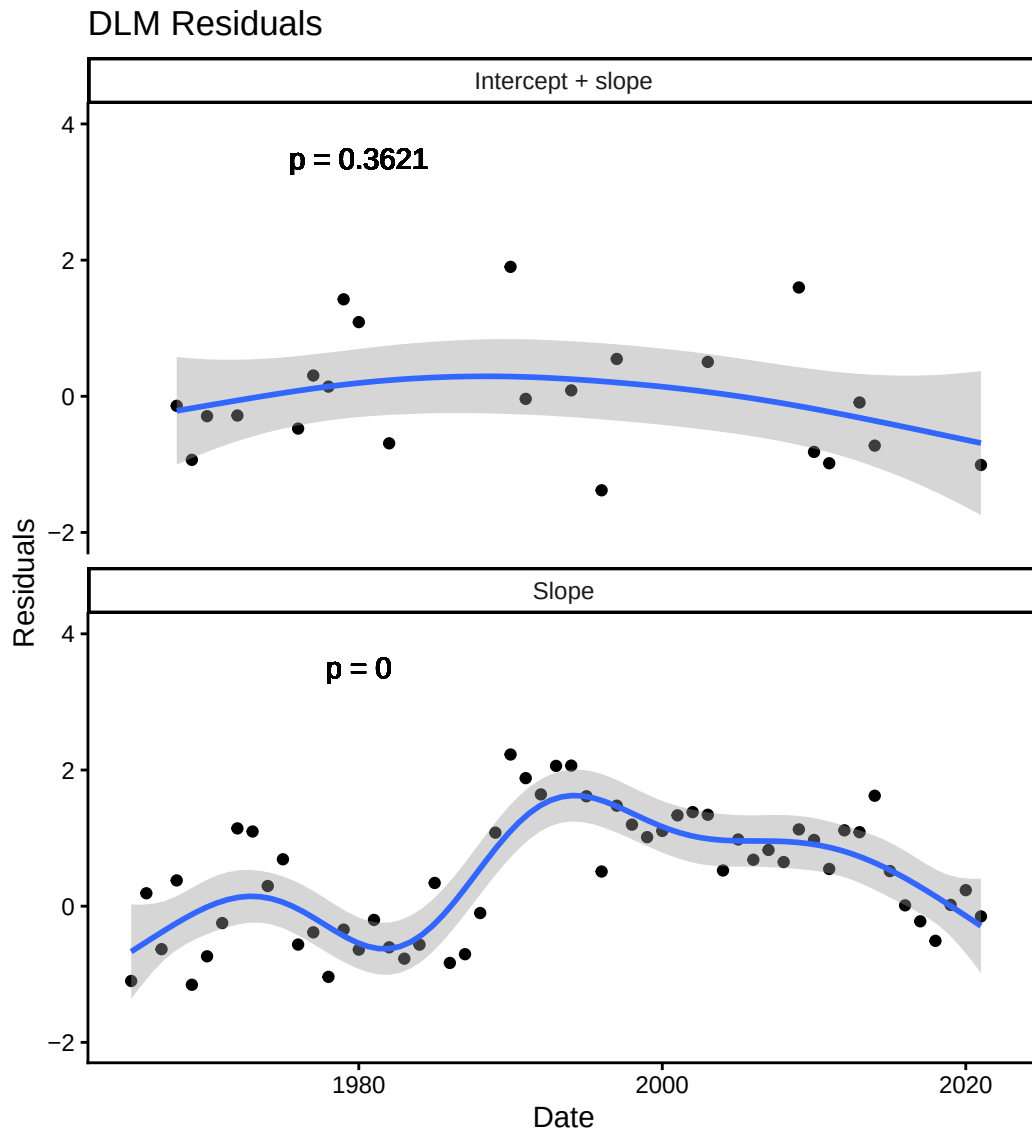


Figure S3.12: Time series of residuals from the DLM model fit with a time-dependent slope and both a time-dependent slope and intercept. p-value denotes the significance of the trend in residuals through time.

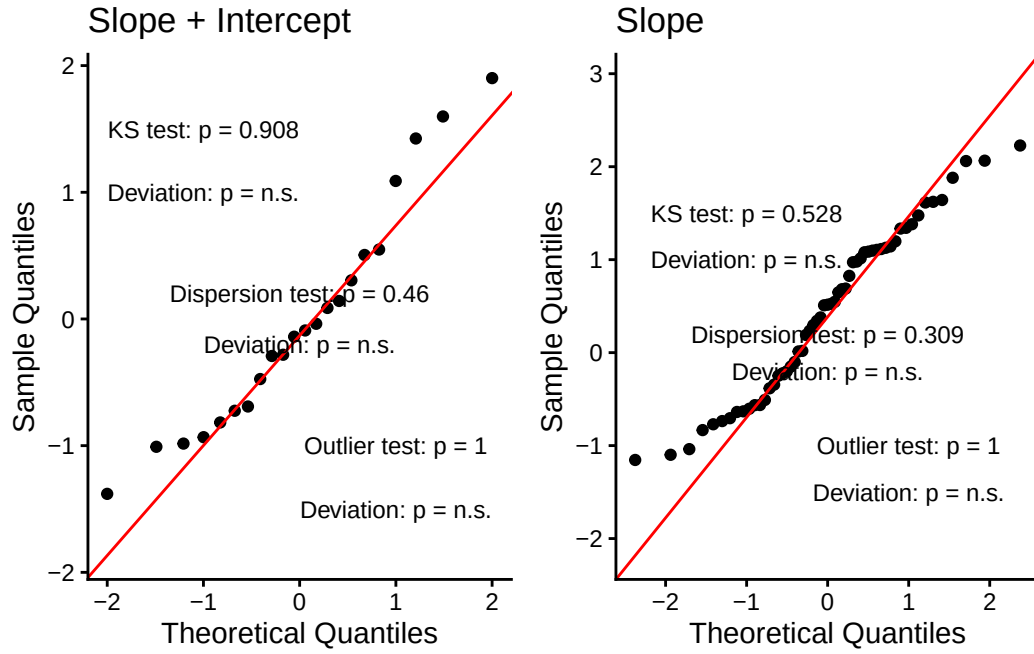


Figure S3.13: QQ plot from the DLM model fit with a time-dependent slope and both a time-dependent slope and intercept.

Model	AIC
1. Intercept only	289.46
2. Intercept by period only (time-dependent intercept)	251.43
3. Intercept by period and SST	248.80
4. Interaction between SST and period (time-dependent slope and intercept)	179.02
5. Time-dependent slope	247.52

Table S3.1: AIC values for five linear models

term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.40	0.23	-5.96	0
z_sst	2.78	0.24	11.49	0
as.factor(period)late	3.29	0.30	10.89	0
z_sst:as.factor(period)late	-3.53	0.31	-11.50	0

Table S3.2: Summary of the output of the best supported model (model 4)

	leave-future-out train	leave-future-out test
Time-dependent slope & intercept	0.42	1.03
Time-dependent slope	1.19	1.60

Table S3.3: RMSE for LFO-CV for both test and training datasets for the time-dependent slope and intercept, and the time-dependent slope models.

Appendix 4: DLM Simulations

Appendix 4: DLM Simulations

Overview

The simulation code for this comparison is in [simulations.R](#). Complete code for this file can be found [here](#). As a brief overview, we are interested in understanding what factors of a DLM contribute to models with time-dependent slopes not performing well - specifically, we found with the GoA salmon dataset that fitting both a time-dependent slope and a time-dependent intercept had convergence issues. All of these models are constructed using the MARSS R package, and we are only considering models with time-dependent intercepts and slopes here.

Parameters

We investigated the following time series properties in these simulations:

- Time series length - we compare time series with 25, 50, 100, and 200 time steps
- Covariate time series autocorrelation - we investigated autocorrelation values of -0.9, -0.5, 0, 0.5, and 0.9. This autocorrelation represents the underlying autocorrelation in the covariate (driver or predictor) variable, not the autocorrelation in the random walk of the intercept or slope
- Covariate trend - after imposing autocorrelation, we applied a linear trend in the covariate, using values of 0 (no trend), 0.01, and 0.02
- Covariate variability - as most applications standardize covariate time series prior to model fitting, we fixed the standard deviation of the covariate time series at 1.0
- Standard deviation of the random walk of the intercept parameter - we used a fixed value of 0.1, and only adjusted the standard deviation of the slope (below)

- Standard deviation of the random walk of the slope parameter - we used values of 0.01, 0.1, and 1.0 to control the variability in the underlying slope. These values represent cases where the slope parameter has smaller, equal, and larger variability compared to the intercept.
- Observation error variability - observation / measurement error was added to each simulated time series, and we fixed the standard deviation of this error at a small value (0.01)

Modeling

For each combination of parameters above, we generated 200 simulated time series (using fixed random number seeds). Random walks in the intercept, slope, and covariate time series were all simulated independently, and then combined to generate a latent response time series. Finally, measurement error was added to create observed time series.

For each simulated dataset, we used the **MARSS** package in R; for fast fitting we used the **TMB** estimation (method = “marssTMB”). We generated a number of summary statistics; because we are focused here on the ability to recover trends in the slope parameters, we focus here on the correlation between the estimated time-dependent slope trend and true time-dependent slope trend.

Results

Our results highlight that two parameters in our simulations are largely responsible for cases when DLMS are not able to recover underlying processes: time series length, and ratio of variability in the intercept versus slope.

Each facet here represents a different standard deviation of the random walk in the slope; as we fixed the standard deviation of the intercept random walk at 0.1, the standard deviation in the right facet represents a case where most of the variability is driven by variability in the slope (Figure S4.1). In contrast the leftmost facet represents a scenario where variability is dominated by the intercept. The results highlight that recovering a time-dependent slope trend is very difficult in time series dominated by variability in the intercept, even when time series are long. More data (longer time series) generally increases the ability to recover trends in all cases, regardless of whether variability is dominated by the intercept or slope.

We see a weak effect of covariate autocorrelation; focusing on time series with 100 time steps and situations where variability in the intercept and slopes are equal (standard deviations = 0.1), covariate time series with strong negative autocorrelation result in a slight increase in the ability to recover underlying slope trends (Figure S4.2).

We also see little effect of a linear trend in the covariate; some of the scenarios explored here (such as the case with an autocorrelation of 0 above) include non-stationary trends, so this result is not surprising (Figure S4.3).

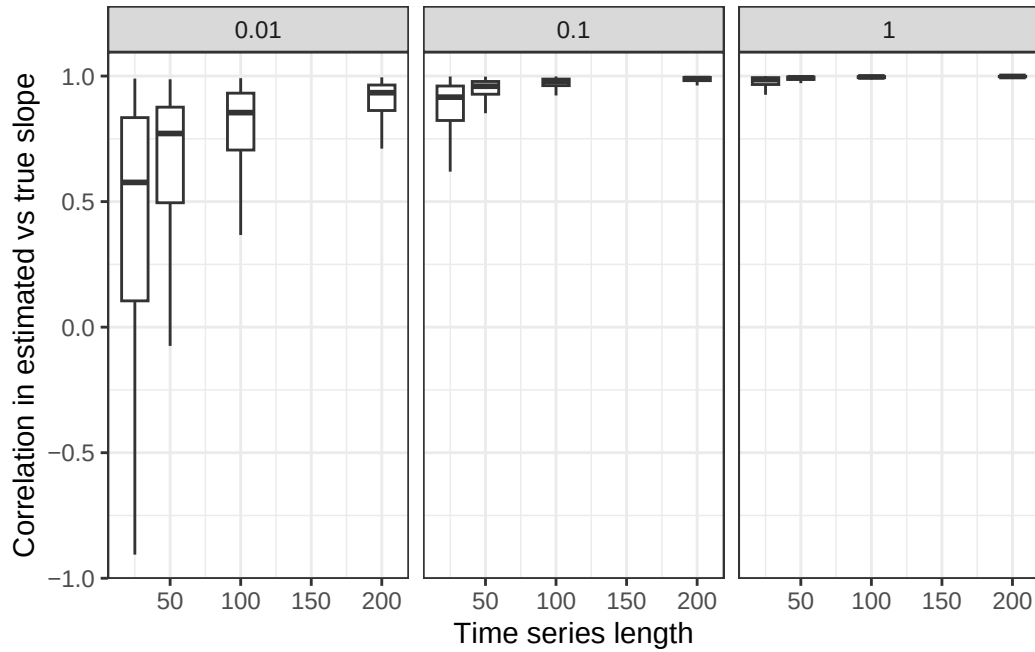


Figure S4.1: Correlation between the model estimated slope and the true slope as a function of time series length. Each panel represents a different ratio in variability between the slope and intercept.

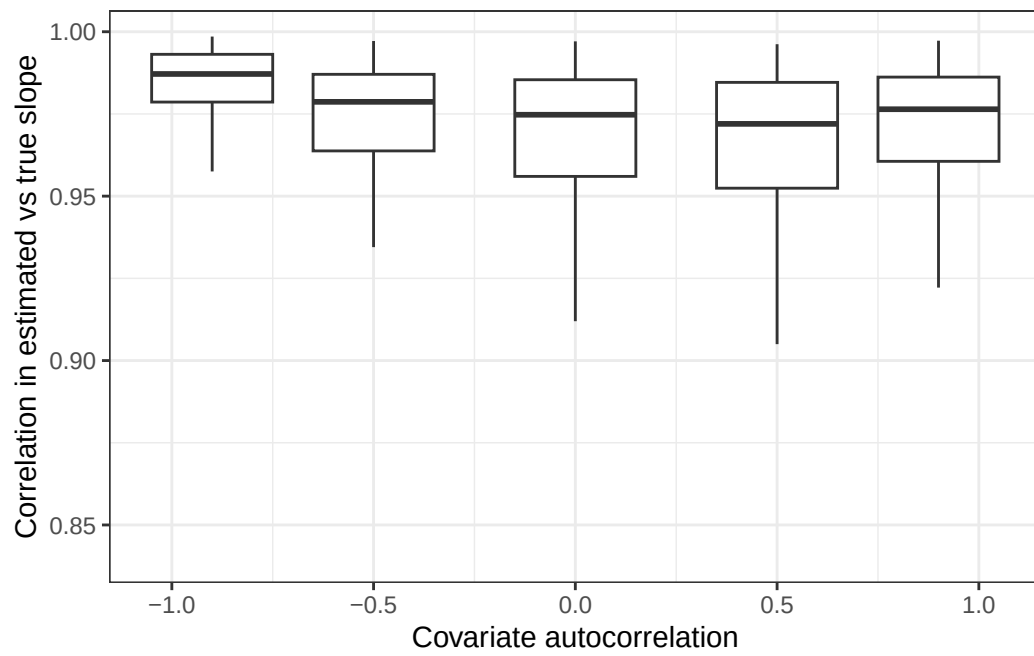


Figure S4.2: Correlation between the model estimated slope and the true slope as a function of autocorrelation

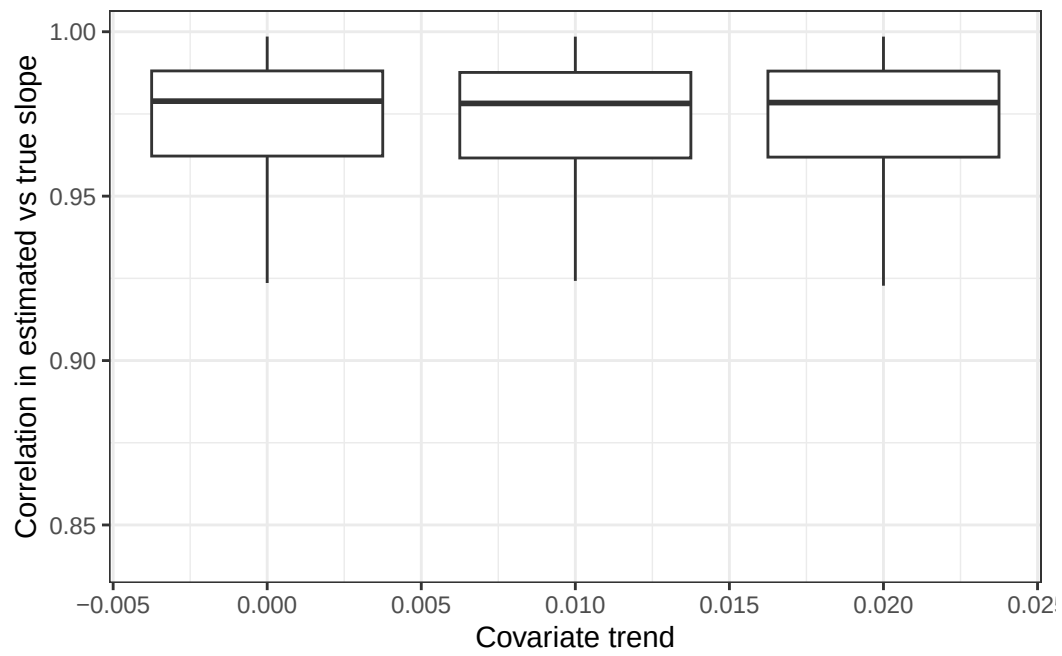


Figure S4.3: Correlation between the model estimated slope and the true slope as a function of the covariate trend